1MFCLA

Mathematical Foundations, Calculus and Linear Algebra

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Lecture Notes – Calculus

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► Learning Outcomes ◄

By the end of this course you should be able to do the following:

- Explain and apply fundamental set-theoretic and functional concepts, including domain, codomain, image and pre-image, injectivity, surjectivity, bijectivity, and cardinality of finite and infinite sets;
- State precise definitions of convergence of a sequence, and of the limit and continuity of a real function, and use them to determine whether a given sequence or function converges, tends to infinity, or is continuous at a point;
- Calculate derivatives of elementary functions directly from the definition of the derivative and by applying the standard rules of differentiation (linearity, product, quotient, chain, inverse-function, implicit, and logarithmic differentiation);
- Evaluate definite and indefinite integrals of elementary functions using the Fundamental Theorem of Calculus together with the techniques of integration by parts, substitution, and partial fractions;
- Determine the convergence or divergence of improper integrals over unbounded intervals and at points of discontinuity, and evaluate them when convergent;
- Apply L'Hôpital's Rule, and related limit techniques, to evaluate limits of indeterminate form;
- Use differentiation to locate and classify stationary points, determine global and local extrema, and sketch the graph of a real function;
- Compute partial derivatives, directional derivatives, and line integrals for simple functions of several variables.

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Chapter 1

Introduction

1.1 Basic concepts

We begin by recalling some basic mathematical concepts and notation for sets and functions. Some of these concepts may already be familiar to you, but the terminology and notation could be different. In any case, it is always good to start on the same page and these introductory concepts will be useful throughout your entire degree.

1.1.1 Sets

Informally speaking, a *set* is a (finite or infinite) collection of objects, called *elements* of the set. Elements can be literally anything meaning that, as we shall see soon, we could talk about a set of numbers, but it would be equally valid to consider a set of elephants. Perhaps even more abstractly, we could also take the elements of a set to be sets themselves, and consider a "set of sets". This is usually fine as well, but one does need to be slightly careful to make sure that the definitions you use to include elements in your set are free of contradictions, so as not to end up in a scenario known as Russell's paradox[1]. We use the symbol $\emptyset$ to denote the *empty set*, that is, the set that has no elements. If an element a belongs to a set A, we write $a \in A$; and if not, we write $a \notin A$.

We will often use different ways to specify a set. For example, we may say "let P be the set of prime numbers". Or, we may list elements either explicitly or implicitly using curly bracket notation, for example $A = \{1, 2, 3\}$ for the set of numbers $1, 2, 3$ or $P = \{2, 3, 5, 7, 11, 13, \ldots\}$ where it is clear P is again the set of prime numbers. We denote by $\mathbb{R}$ the set of real numbers, $\mathbb{Q}$ the set of rational numbers, $\mathbb{Z}$ the set of integers, and $\mathbb{N} = \{1, 2, \ldots\}$ the set of natural numbers. We can also specify a set using set builder

[1] One example of this is the Barber's paradox https://en.wikipedia.org/wiki/Russell's_paradox

notation, for example, we may also write $P = \{n \in \mathbb{N} : n \text{ is prime}\}$ to again denote the set of prime numbers, which reads "P is the set of all n in $\mathbb{N}$ such that n is prime".

Now, let A, B denote two sets. We say that:

- A is a subset of B (perhaps equal to B), denoted $A \subseteq B$, if every element of A is an element of B;
- A equals B, denoted $A = B$, if A and B have exactly the same elements (note this implies $A \subseteq B$ and $B \subseteq A$);
- A is a proper subset of B, denoted $A \subset B$, if $A \subseteq B$ but A is not equal to B (denoted $A \neq B$).

Example 1.1. $A = \{2, 5\}$ is a proper subset of P, the set of prime numbers. Thus, we write $A \subset P$.

There is also an algebra of sets. For this, let $A, B \subseteq X$ be two subsets of a (possibly larger) set X. We then define:

- The *union* of A and B, denoted $A \cup B$, is the set defined by
$$A \cup B = \{x \in X \ : \ x \in A \quad \text{or} \quad x \in B\}.$$
- The *intersection* of A and B, denoted $A \cap B$, is the set defined by
$$A \cap B = \{x \in X \ : \ x \in A \quad \text{and} \quad x \in B\}.$$
If $A \cap B = \emptyset$, then A and B have no elements in common and A and B are said to be *disjoint*.
- The *complement of A in X*, denoted A^c (or sometimes $X \setminus A$) is the set
$$A^c = \{x \in X : x \notin A\}.$$

Example 1.2. Let $A = \{1, 2, 3, 4, 5\}$ and P be the set of prime numbers. Then

$$A \cup P = \{1, 2, 3, 4, 5, 7, 11, 13, ...\}$$

where it is understood that all numbers to the right are prime, and

$$A \cap P = \{2, 3, 5\}.$$

The complement of P in $\mathbb{N}$ can be written $P^c = \{n \in \mathbb{N} : n \text{ is not prime}\}$.

Observe that, for the definition of union and intersection it is not really necessary to involve the additional set X, but for the definition of complement there must be a "superset" X from which to "remove" elements of A. Hence, the notation $X \setminus A$ for set difference is commonly used in place of A^c unless it is implicitly understood that the latter means "complement in X". The notion of an algebra of sets follows since these operations as defined satisfy the following properties. Specifically, let A, B and C be sets. Then, it can be shown (left as an exercise to prove) that:

- **(Transitivity of $\subseteq$)** If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.
- **(Associativity)** The operations of union and intersection individually satisfy
$$(A \cup B) \cup C = A \cup (B \cup C),$$
and
$$(A \cap B) \cap C = A \cap (B \cap C).$$
- **(Distributivity)** The operations of union and intersection together satisfy
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$$
and
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$
- **(De Morgan's Laws)** The set difference satisfies:
$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C),$$
and
$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$$

In terms of notation A^c for complements, De Morgan's Laws can equivalently be written

$$(A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (A \cap B)^c = A^c \cup B^c.$$

It helps to draw some pictures to gain intuition for this result, as shown in Figure 1.1.

Finally, we introduce the concept of an ordered pair and the Cartesian product of sets. Unlike set notation where, given $a \in A, b \in B$ we can write $\{a, b\} = \{b, a\}$ (the order of elements in a set does not matter), we say that (a, b) is an *ordered pair* if, whenever $a, a' \in A$, $b, b' \in B$ and $(a, b) = (a', b')$, then $a = a'$ and $b = b'$. In words an ordered pair (a, b) is a pair of elements in which their order is significant. Cartesian products and binary relations (and hence functions, as we will see below) are defined in terms of ordered pairs. Let A, B be sets. Then, the *Cartesian product* of A and B, denoted $A \times B$, is defined to be the set of all ordered pairs whose first entry (or coordinate) is in A and the second in B:

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

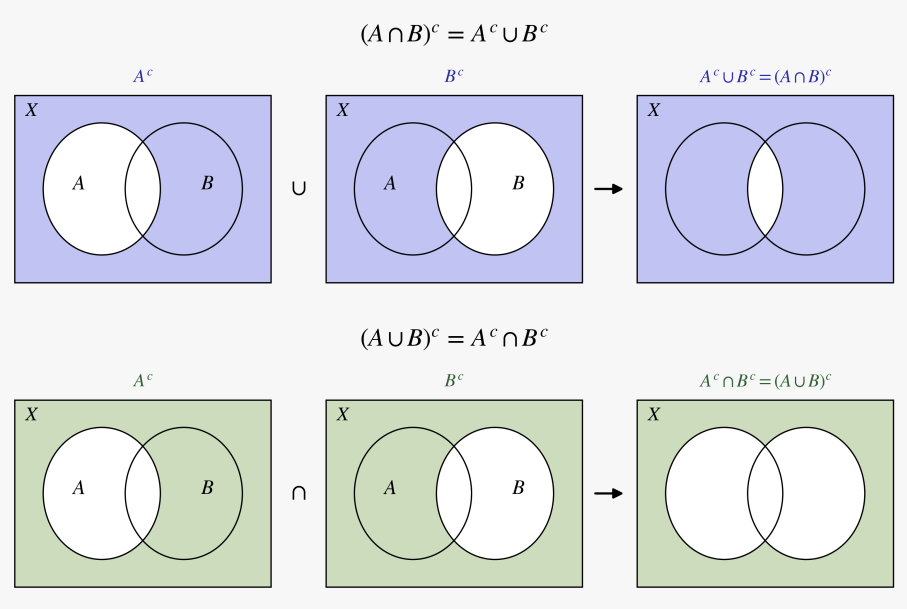


Figure 1.1: Diagram illustrating De Morgan's Laws, involving subsets $A, B \subset X$ and their complements in X. The shaded regions denote the corresponding sets.

1.1.2 Functions

Previously, we introduced ordered pairs and said that they could be used to define binary relations and thus functions. Indeed, this is the most formal (abstract) way to define functions but we will not do that here. Instead, for our purposes a function f mapping from a set X to another set Y, denoted $f : X \to Y$, is defined by three properties:

- a set $X = \text{dom}(f)$ called the *domain* of f;
- a set $Y = \text{codom}(f)$ called the *codomain* of f;
- a *well-defined rule* that assigns a *unique element* $f(x) \in Y$ to each element $x \in X$.

When talking about functions, we often call the elements of X and Y *points*. We say that a rule f assigning elements of X to elements of Y is well-defined or unambiguous provided that, whenever $x_1 = x_2 \in X$ we have $f(x_1) = f(x_2) \in Y$. That is, we want a function from X to Y to give a unique output for each element of X, that output should be in Y, and its value should be unambiguously defined. It turns out that binary relations do not need to satisfy the uniqueness property.

In my experience, students can sometimes get confused about the concepts of domain and codomain of a function. I believe the source of confusion regarding domain partly stems from thinking about a function only as f, rather than the function being specified

together with domain and codomain, as in $f : X \to Y$. When written this way it is clear that f must be defined on all elements or points of the domain X. Otherwise, the definition of $f : X \to Y$ does not make sense.

Example 1.3. The function $f(x) = 1/x$ that takes the reciprocal of a real number is not defined at the point $x = 0$, since everybody knows that $1/0$ is not well-defined. Thus, rather than the domain of f being the set of real numbers $\mathbb{R}$, in this case the "natural domain" of f is $\mathbb{R}\setminus\{0\}$, the set of real numbers with the point 0 removed. So, we can write $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ given by $f(x) = 1/x$, where it is clear the function is now well-defined.

Unfortunately, people do not always write a function involving its domain and codomain this way and so often in calculus or mathematics in general, we will have to agree that a function is defined implicitly on its "natural domain" (the largest domain where the function is defined in the working context) unless otherwise specified.

A proper grasp of the concept of codomain turns out to be critical for understanding the different types of functions that we shall see, which also involves the definition of the image of a function. Let $f : X \to Y$. The *image* of f is the set of all outputs across the entire domain, denoted

$$\mathrm{im}(f) = \{f(x) : x \in X\}.$$

More generally, if $A \subseteq X$ is a subset of the domain X, the *image of A under f* is

$$f(A) = \{f(x) : x \in A\},$$

which is the set of all outputs where f is applied to the subset A. While the domain of $f : X \to Y$ is the set of all input values for f (so that, as argued above, X should not include elements where the function is not defined), the codomain is the set of possible outputs. Notice that not every element of the codomain needs to be an output of the function.

Example 1.4. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$. Clearly no negative number, e.g. -1, contained within codomain $\mathbb{R}$ can be found equal to the output $f(x)$ of any x in the domain $\mathbb{R}$. Thus, in this case the image of the function is the set of non-negative real numbers, often denoted $\mathbb{R}_{\geq 0} = \{x \in \mathbb{R} : x \geq 0\}$.

So we see that the codomain Y of $f : X \to Y$ need not be equal to the image of f, yet it remains well-defined.

Suppose that you are given $f : X \to Y$. A natural question is to ask which values in the domain X give rise to a particular point $y \in Y$ in the codomain. Or, more generally, given a subset of the codomain $B \subseteq Y$, we may ask which values of the domain gives rise to points in B. If $B \subseteq Y$, the *pre-image* of B is

$$f^{-1}(B) = \{x \in X : f(x) \in B\}.$$

Example 1.5. From our previous example of $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$, we see that the pre-image of $\mathbb{R}_{\geq 0} \subset \mathbb{R}$ is the entire domain $f^{-1}(\mathbb{R}_{\geq 0}) = \mathbb{R}$. The pre-image of a subset containing only one strictly positive number, say $+1 \in \mathbb{R}$, is $f^{-1}(\{+1\}) = \{-1, +1\}$, the set containing the two square roots of that number. The pre-image of a subset containing only one strictly negative number, say $-1 \in \mathbb{R}$, is always the empty set $f^{-1}(\{-1\}) = \emptyset$, and the pre-image of the subset containing only zero is the set containing only zero $f^{-1}(\{0\}) = \{0\}$.

By slight abuse of notation, people sometimes write $f^{-1}(y)$ for the pre-image when $B = \{y\}$ is a subset of the codomain containing just a single point, and this can often be confused with the inverse of a function, which is a different concept we shall see below.

Different types of functions can be defined based on how the mapping rule interacts with the domain, image and codomain. Suppose that $f : X \to Y$. Then, we define:

- **(Surjection)** We say that f is *surjective* (or *onto*) if, for every $y \in Y$, there exists at least one $x \in X$ such that $f(x) = y$.
- **(Injection)** We say that f is *injective* (or *one-to-one*) if, for every $y \in Y$, there is at most one $x \in X$ such that $f(x) = y$.
- **(Bijection)** We say that f is *bijective* if f is both surjective and injective.

These definitions often require a bit of thinking about and sketching out to fully appreciate, see Figure 1.2. They also distil the requirement to define a function in the context of domain and codomain.

If $f : X \to Y$ is a surjection, this equivalently means that every point in the codomain is also in the image, so that $\text{im}(f) = \text{codom}(f) = Y$ if and only if $f : X \to Y$ is surjective.

Example 1.6. The function $f : \mathbb{R} \to \mathbb{R}_{\geq 0}$ given by $f(x) = x^2$ is a surjection since every non-negative number has a real square root, but $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ is not surjective, simply because the codomain has changed.

If $f : X \to Y$ is an injection, this equivalently means distinct elements of the domain are mapped to distinct elements in the codomain; namely, if $x_1, x_2 \in X$ are two points in the domain then $f(x_1) = f(x_2)$ implies $x_1 = x_2$.

Example 1.7. The function $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ given by $f(x) = x^2$ is an injection since every non-negative real number has a unique non-negative square root, but $f : \mathbb{R} \to \mathbb{R}_{\geq 0}$ given by $f(x) = x^2$ is not injective, simply because the domain has changed.

There are clearly examples of functions that are neither surjective nor injective.

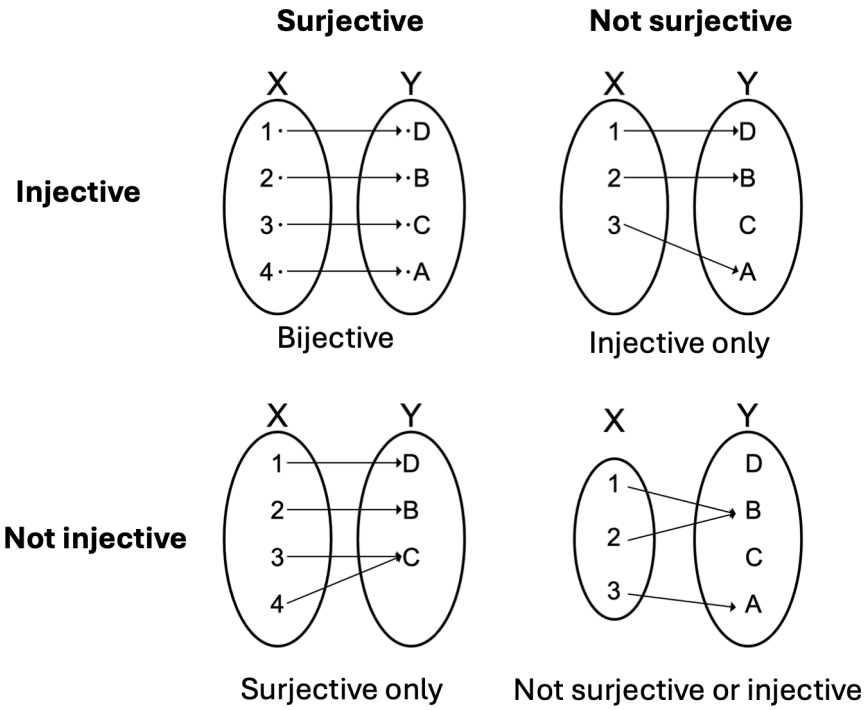


Figure 1.2: Diagram illustrating the different types of functions, with examples represented by arrows mapping elements of domain X to elements of codomain Y. Columns split examples by whether the function is surjective or not, while rows split examples by whether the function is injective or not. Top left gives an example of a bijective function (both surjective and injective), off diagonals are examples of functions that are only surjective or injective, and bottom right gives an example of a function that is neither. Adapted from Wikipedia, Original: Schapel; vectorized by Samuel Mediani.

Example 1.8. The function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ shown not to be surjective above is actually not injective either, since every strictly positive real number has both a positive and a negative real square root.

Bijective functions, which are both surjective and injective, are particularly special since the inverse of a function exists if and only if it is a bijection. Indeed, let $f : X \to Y$ be a bijective function. Then the *inverse* of f, is the function $f^{-1} : Y \to X$ defined by the rule $f^{-1}(y) = x$ if and only if $f(x) = y$.

Example 1.9. The function $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ given by $f(x) = x^2$ is both injective (as shown in previous example) and surjective, and thus bijective, since every non-negative real number in the codomain has a unique non-negative square root. The inverse of this function is $f^{-1} : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ given by $f^{-1}(y) = \sqrt{y}$, taking the non-negative square root.

Note that we use the notation f^{-1} in two different ways, here to denote the inverse of a bijective function and previously to denote the pre-image. It is important to remember that the latter exists and is well-defined for *any* function, while only bijective functions have an inverse.

Finally, we consider a general way to combine two or more functions to produce another. Let $f : X \to Y$ and $g : Y \to Z$ be two functions. The *composition* of g with f, denoted $g \circ f : X \to Z$, is the function given by $(g \circ f)(x) = g(f(x))$.

Example 1.10. Let $f : \mathbb{R} \to \mathbb{R}$ is given by $f(x) = x^2$ and $g : \mathbb{R} \to \mathbb{R}$ be given by $g(x) = x^3$. Then, $g \circ f : \mathbb{R} \to \mathbb{R}$ is given by $(g \circ f)(x) = g(x^2) = (x^2)^3 = x^6$.

Note that the composition of f with g is not necessarily the same as the composition of g with f, since in general the domains and codomains have to match: if we want to consider $f \circ g : Y \to Z$ then we must have $f : X \to Z$ and $g : Y \to X$. A special case is obtained by composing a bijective function with its inverse. Then we find that $f^{-1} \circ f : X \to X$ is given simply by $(f^{-1} \circ f)(x) = f^{-1}(f(x)) = x$, which is also known as the *identity function* on X, often denoted $\mathrm{id}_X : X \to X$ with $\mathrm{id}_X(x) = x$.

1.1.3 Cardinality and countability

For finite sets, the notion of the size of a set is naturally identified with the number of elements within it, but it seems more difficult to ascribe "sizes" to infinite sets. However, the size or cardinality of a set can also be understood through the existence of injections or bijections and it turns out this idea extends naturally to infinite sets. The *cardinality* of a set X, denoted $|X|$, is an attribute of X such that for any sets X and Y:

- $|X| \leq |Y|$ if and only if there exists an injection $f : X \to Y$;

- $|X| = |Y|$ if and only if there exists a bijection $f : X \to Y$.

Thus, cardinality is measured by putting sets in one-to-one correspondence: each element in X can be assigned a dedicated "pair" in Y, and no element from either set left unpaired. If this is possible, the sets are said to have the same cardinality, and if not, one set is said to be strictly larger or strictly smaller than the other. In particular, this definition of cardinality justifies our ability to write a finite set X of size n as $X = \{x_1, ..., x_n\}$ where $x_1, ..., x_n$ are all distinct. Indeed, in doing so we implicitly choose a bijection $f : \{1, 2, ..., n\} \to X$ (whose existence is guaranteed by the definition of size) and set $x_i = f(i)$ for each $i \in \{1, 2, ..., n\}$.

Now, the same concepts extend to infinite sets X, Y where, to show $|X| \leq |Y|$, it suffices to find an injective function $f : X \to Y$. To show $|X| = |Y|$, it suffices to find a bijection $f : X \to Y$. Interestingly, a proper subset of an infinite set X can have the same cardinality as X.

Example 1.11. Using $2\mathbb{Z} \subset \mathbb{Z}$ to denote the set of even integers, which is a proper subset of the set of integers $\mathbb{Z}$, we have a simple bijection $f : \mathbb{Z} \to 2\mathbb{Z}$ given by $f(n) = 2n$. This means that $|2\mathbb{Z}| = |\mathbb{Z}|$, the cardinality or "size" of the set of even integers is the same as the set of integers!

Things don't get too crazy, however: for any (possibly infinite) subset $X \subseteq Y$, we can always form the injective function $f : X \to Y$ given by $f(x) = x$, which shows we must have $|X| \leq |Y|$, i.e. cardinality of a subset can never be larger than that of the superset.

One particularly important cardinality is $|\mathbb{N}|$, the cardinality of the set of natural numbers. Since this set contains an infinite number of elements, one may be inclined to say its cardinality is "infinity", and in some sense that is true, but it is certainly not the same infinity we will come to understand much better in this calculus course. We have seen how to consider cardinalities using injections and bijections, so we may be better off asking: are there sets with cardinalities that are smaller or larger than $|\mathbb{N}|$? One of the reasons that $|\mathbb{N}|$ is important is that it has the smallest cardinality an infinite set can have, and for that reason we introduce special notation $\aleph_0 = |\mathbb{N}|$. Indeed, let X be a set. Then either X is finite or $\aleph_0 \leq |X|$. We will not prove this result here.

Next, we say a set X is *countable* if either it is finite or if there exists a bijection $f : \mathbb{N} \to X$, which in the second case implies $|X| = \aleph_0$. In the second case, we say that X is *countably infinite*. If a set X is not countable then we say it is *uncountable*. It may help to think of countably infinite sets as being "small" infinite sets, and uncountable sets as being "large" infinite sets. Indeed, countable sets are those which we can "count" using the natural numbers.

Example 1.12. A bijection $f : \mathbb{N} \to \mathbb{Z}$ is given by

$$f(n) = \begin{cases} \dfrac{n-1}{2}, & \text{if } n \text{ is odd,} \\ -\dfrac{n}{2}, & \text{if } n \text{ is even.} \end{cases}$$

Hence the natural numbers and integers have the same cardinality, $|\mathbb{Z}| = \aleph_0$, or, equivalently, the integers are countable. Based on our previous example, we therefore see that $2\mathbb{Z}$ is countable as well.

The next result is perhaps surprising. There is a cute way to show that the set $\mathbb{N} \times \mathbb{N}$ is also countable by writing out the elements of $\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$ on the same infinite grid as follows:

$(1,1)$	$(1,2)$	$(1,3)$	$(1,4)$	$\cdots$
$(2,1)$	$(2,2)$	$(2,3)$	$(2,4)$	$\cdots$
$(3,1)$	$(3,2)$	$(3,3)$	$(3,4)$	$\cdots$
$(4,1)$	$(4,2)$	$(4,3)$	$(4,4)$	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	

1	3	6	10	$\cdots$
2	5	9	14	$\cdots$
4	8	13	19	$\cdots$
7	12	18	25	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	

The right-hand side shows how the elements of this grid may be labelled with elements of $\mathbb{N}$, by proceeding up each diagonal in turn from bottom left to top right. This creates a bijection from $\mathbb{N}$ to $\mathbb{N} \times \mathbb{N}$, namely the function $f : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$ in which each $n \in \mathbb{N}$ in the right-hand grid maps to the ordered pair $(m, p) \in \mathbb{N} \times \mathbb{N}$ in the corresponding position in the left-hand grid (so, for instance, $f(10) = (1, 4)$ and $f(8) = (3, 2)$). Indeed, f is well-defined since every natural number appears once in the right-hand grid and so is mapped to exactly one pair in the left-hand grid; f is injective and surjective since each pair appears precisely once in the left-hand grid so is mapped to by exactly one element of $\mathbb{N}$. The existence of this bijection proves that $\mathbb{N} \times \mathbb{N}$ is countably infinite. It can also be shown that the set of rational numbers $\mathbb{Q}$ is countably infinite.

Finally, you may be left wondering which sets are in fact uncountable. It turns out that the set of real numbers $\mathbb{R}$ is uncountable. Proof of this statement follows an argument first put forward by Cantor[2], which involves showing that $\{0, 1\}^{\aleph_0}$, the set of all infinite sequences of zeros and ones, is uncountable and then identifying a bijection $f : \{0, 1\}^{\aleph_0} \to \mathbb{R}$. Actually, Cantor demonstrated that there is no largest cardinality: for every set X we can obtain a strictly larger set by taking the power set $\mathcal{P}(X) = \{A \subseteq X\}$, whose elements are the subsets of X, and the cardinality $|\mathcal{P}(X)|$ is always strictly larger than $|X|$. Thus, starting from a countably infinite set, one can repeatedly take power sets to obtain an infinite sequence of ever-larger cardinalities. It also turns out that the cardinality $|\mathbb{R}|$ is equal to $|\mathcal{P}(\mathbb{N})|$, which is denoted $2^{\aleph_0}$. More generally, set theory organises infinite cardinalities into a hierarchy of aleph numbers $\aleph_0, \aleph_1, \aleph_2, \ldots$ used to represent the cardinality of special infinite sets. The cardinalities obtained by repeated power-set operations do not in

[2]Georg Cantor played a pivotal role in the creation of set theory https://en.wikipedia.org/wiki/Georg_Cantor

general coincide with successive aleph numbers, however. The unanswerable Continuum Hypothesis[3] posits $\aleph_1 = 2^{\aleph_0}$. For now, we will stick with calculus over the real numbers where a tamer version of "infinity" makes much more sense!

1.2 Sets and functions of real numbers

We now turn to focus specifically on sets and functions involving real numbers, which serves as a basis for the study of calculus. Calculus shares many overlaps with real analysis, which tends to develop the study of functions over the real numbers through a more formal, proof-based approach. By comparison, calculus is typically viewed as a collection of tools for doing computations with real functions. You may or may not be taking a real analysis course during your degree and therefore this material will provide enough background required for both calculus and an initial primer for more formal treatment of the real numbers.

1.2.1 Modulus

One of the most important functions of real numbers is the modulus. The *modulus* or *absolute value* of a real number $x \in \mathbb{R}$ is defined by

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

We read $|x|$ as "the modulus of x" or "the absolute value of x".

Let $x, y \in \mathbb{R}$. There are several fundamental properties of the modulus:

- $0 \leq |x|$ and $|-x| = |x|$;
- $-|x| \leq x \leq |x|$;
- $|x| \leq y$ if and only if $-y \leq x \leq y$;
- $|xy| = |x||y|$;
- **(Triangle Inequality)** $|x + y| \leq |x| + |y|$;
- **(Reverse Triangle Inequality)** $\big||x| - |y|\big| \leq |x - y|$.

[3]This hypothesis cannot be answered within the standard axioms of mathematics https://en.wikipedia.org/wiki/Continuum_hypothesis

The triangle inequality is one of the most important properties. To prove it, we consider the two cases:

Case 1: $x + y \geq 0$. Then $|x + y| = x + y$ and since $x \leq |x|$ and $y \leq |y|$, we obtain

$$|x + y| = x + y \leq |x| + |y|.$$

Case 2: $x + y < 0$. Then

$$|x + y| = -(x + y) = (-x) + (-y),$$

and since $-x \leq |-x| = |x|$ and $-y \leq |-y| = |y|$, it follows that

$$|x + y| = (-x) + (-y) \leq |x| + |y|.$$

Hence, in both cases, $|x + y| \leq |x| + |y|$, as required.

For $x, y, z \in \mathbb{R}$, the triangle inequality immediately gives the related forms

$$|x - y| = |x + (-y)| \leq |x| + |y|,$$

and

$$|x - y| = |(x - z) + (z - y)| \leq |x - z| + |z - y|.$$

The latter means $|x - y|$ is often interpreted geometrically as the Euclidean distance between $x, y \in \mathbb{R}$ on the real line, which you will see more about if you are taking a real analysis course.

1.2.2 Intervals and bounded sets

We now consider some fundamental subsets of the real numbers. An *interval* is a subset of $\mathbb{R}$ in one of the following forms, where $a, b \in \mathbb{R}$ and $a < b$:

$$\begin{aligned}
(a, b) &:= \{x \in \mathbb{R} : a < x < b\} && \text{(this is called an } \textit{open} \text{ interval)} \\
(a, b] &:= \{x \in \mathbb{R} : a < x \leq b\} \\
[a, b) &:= \{x \in \mathbb{R} : a \leq x < b\} \\
[a, b] &:= \{x \in \mathbb{R} : a \leq x \leq b\} && \text{(this is called a } \textit{closed} \text{ interval)} \\
(a, \infty) &:= \{x \in \mathbb{R} : x > a\} \\
[a, \infty) &:= \{x \in \mathbb{R} : x \geq a\} \\
(-\infty, b) &:= \{x \in \mathbb{R} : x < b\} \\
(-\infty, b] &:= \{x \in \mathbb{R} : x \leq b\} \\
(-\infty, \infty) &:= \mathbb{R}
\end{aligned}$$

Observe that this convention serves as an alternative to expressing subsets using inequalities. In this notation, a square bracket means the corresponding endpoint is included

(think $a \leq x$), while a round bracket means the endpoint is excluded (think $a < x$). For example, $[0, \infty)$ contains the point 0 while $(0, \infty)$ does not. Note that "∞" is not a number and so it does not make sense to write $[0, \infty]$ as a subset of $\mathbb{R}$. Also, note that this definition excludes singleton sets such as $\{a\}$, which could be written as $[a, a]$, but this is simply a matter of convention and we could consider these to be intervals as well.

We can combine intervals using the set operations union and intersection to create new subsets of $\mathbb{R}$.

Example 1.13. The union of $(-\infty, 0)$ and $(0, \infty)$ gives a subset we saw previously:

$$\mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty).$$

On the other hand, the intersection of $(-\infty, 0]$ and $[0, \infty)$ contains only the point 0:

$$\{0\} = (-\infty, 0] \cap [0, \infty).$$

We can often use this concept to simplify set builder notation for subsets defined by multiple conditions.

Example 1.14. We have

$$\{x \in \mathbb{R} : x \leq 2, x \neq 1\} = (-\infty, 1) \cup (1, 2].$$

Finally, it will be useful to have a notion of a bounded set of real numbers. A subset $X \subseteq \mathbb{R}$ is called *bounded* if there exists a real number $K \in \mathbb{R}$ such that

$$|x| \leq K \quad \text{for all } x \in X.$$

Note that if $a, b \in \mathbb{R}$ are such that $a < b$, then the intervals (a, b) and $[a, b]$ are bounded sets. Since, for $x \in [a, b]$, we have $|x| \leq \max\{|a|, |b|\}$, we can always take $K = \max\{|a|, |b|\}$. A subset $X \subseteq \mathbb{R}$ for which no such K exists is called *unbounded*. For example, the interval $(0, \infty)$ is unbounded since for every $K \in \mathbb{R}$ there exists $x \in (0, \infty)$ with $|x| > K$.

1.2.3 Solving inequalities

The previous subsection defined subsets of the real numbers using inequalities, which you will likely already feel comfortable manipulating and solving. Here, we recap some of the methods used for solving such inequalities by including our new concepts and notation.

Let us recall how to solve an inequality from the following example.

Example 1.15. Assume that we want to solve the inequality

$$x^3 \geq 2x - 1,$$

that is, determine the set of all the $x \in \mathbb{R}$ that satisfy this inequality.

Solution: It is convenient to move all terms to the left-hand side, thus obtaining the equivalent inequality

$$x^3 - 2x + 1 \geq 0.$$

It is easily observed that $x = 1$ is a root of the cubic polynomial in the left-hand side; this can be used to factorise it and rewrite the inequality as

$$(x - 1)(x^2 + x - 1) \geq 0.$$

After a further factorisation we obtain

$$(x - 1)(x - a)(x - b) \geq 0$$

where roots of $x^2 + x - 1$ are found to be $a = -(1 + \sqrt{5})/2$ and $b = (\sqrt{5} - 1)/2$. Since $a < b < 1$, we can use the following diagram to represent the sign of the individual factors in the left-hand side, and the corresponding sign of their product (on the bottom row).

	$x < a$	$x = a$	$a < x < b$	$x = b$	$b < x < 1$	$x = 1$	$1 < x$
$x - a$	$-$	0	$+$	$+$	$+$	$+$	$+$
$x - b$	$-$	$-$	$-$	0	$+$	$+$	$+$
$x - 1$	$-$	$-$	$-$	$-$	$-$	0	$+$
$x^3 - 2x + 1$	$-$	0	$+$	0	$-$	0	$+$

From the above diagram, it is clear that $x^3 - 2x + 1 \geq 0$ if and only if $a \leq x \leq b$ or $x \geq 1$. In other words, the set of solutions $\{x \in \mathbb{R} : x^3 \geq 2x - 1\}$ can be written by using the interval notation as

$$[a, b] \cup [1, \infty).$$

□

The previous example shows a systematic way to solve inequalities is often through *factorisation* and *sign analysis*.

Whenever a modulus appears in an inequality, the definition

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0, \end{cases}$$

suggests splitting into cases.

Example 1.16. Solve the inequality

$$|x - 1| > x^2 - 2x.$$

Solution:

Case 1: $x < 1$. Then $x - 1 < 0$ and so

$$|x - 1| = -(x - 1) = 1 - x.$$

Hence we must solve $1 - x > x^2 - 2x$, which simplifies to $x^2 - x - 1 < 0$. The roots of $x^2 - x - 1$ are $(1 - \sqrt{5})/2$ and $(1 + \sqrt{5})/2$ and in this case x must lie strictly between these. However, since we are assuming $x < 1$, the solution in this case is

$$x \in \left(\frac{1 - \sqrt{5}}{2}, \frac{1 + \sqrt{5}}{2}\right) \cap (-\infty, 1) = \left(\frac{1 - \sqrt{5}}{2}, 1\right).$$

Case 2: $x \geq 1$. Then $x - 1 \geq 0$ and so

$$|x - 1| = x - 1.$$

Hence we must solve $x - 1 > x^2 - 2x$, which simplifies to $x^2 - 3x + 1 < 0$. The roots of $x^2 - 3x + 1$ are $(3 - \sqrt{5})/2$ and $(3 + \sqrt{5})/2$ and intersecting with $x \geq 1$ gives

$$x \in \left(\frac{3 - \sqrt{5}}{2}, \frac{3 + \sqrt{5}}{2}\right) \cap [1, \infty) = \left[1, \frac{3 + \sqrt{5}}{2}\right).$$

Finally, combining both cases we obtain

$$x \in \left(\frac{1 - \sqrt{5}}{2}, 1\right) \cup \left[1, \frac{3 + \sqrt{5}}{2}\right) = \left(\frac{1 - \sqrt{5}}{2}, \frac{3 + \sqrt{5}}{2}\right).$$

□

The following general approach to solving inequalities is useful in many situations:

1. Rearrange so that one side is zero.
2. Factorise wherever possible.
3. Construct a sign table.
4. Determine where the expression is positive or negative.
5. For modulus expressions, split into cases using the definition of $|x|$.

1.2.4 Elementary functions and domain convention

Elementary functions are functions of a single variable (real-valued functions of a real variable in this module) that are defined by taking sums, products, quotients, inverses, and compositions of finitely many of the following basic functions that are defined on their respective natural domains X:

- **(Constant functions)**: the function $f : X \to \mathbb{R}$ is said to be a *constant* function if there exists some fixed $c \in \mathbb{R}$ such that $f(x) = c$ for all $x \in X$;
- **(Power functions)**: the function $f : X \to \mathbb{R}$ is said to be a *power function with power* α if $f(x) = x^\alpha$ with fixed $\alpha \in \mathbb{R} \setminus \{0\}$ for all $x \in X$, where X is the largest domain where this expression defines a real-valued function; all polynomials can be written as a combination of power functions and constants, e.g.

$$p(x) = 5x^4 + 2x^3 - 5x^2 - 1;$$

- **(Exponential functions)**: the function $f : X \to \mathbb{R}$ is said to be an *exponential function with base* a if $f(x) = a^x$ with fixed $a > 0$, $a \neq 1$ for all $x \in X$; the exponential function e^x (where e denotes Euler's number [4]) or $\exp(x)$ plays a very important role in calculus;
- **(Trigonometric functions)**: These are the functions defined in trigonometry, such as $\sin(x)$, $\cos(x)$, $\tan(x)$, $\cot(x)$, etc., where

$$\tan(x) = \frac{\sin(x)}{\cos(x)}, \quad \cot(x) = \frac{\cos(x)}{\sin(x)};$$

- **(Hyperbolic functions)**: These are the functions $\sinh(x)$, $\cosh(x)$, $\tanh(x)$, $\coth(x)$, etc., defined in terms of exponential functions

$$\sinh(x) = \frac{e^x - e^{-x}}{2}, \quad \cosh(x) = \frac{e^x + e^{-x}}{2},$$

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)}, \quad \coth(x) = \frac{\cosh(x)}{\sinh(x)};$$

- **(Inverses of the above, if they exist)**: The logarithm is the inverse of the exponential (both natural, of base e, and of base a), and inverse trigonometric functions should already be familiar to you; inverses of the hyperbolic functions are denoted similarly:

$$\ln(x), \quad \log_a(x), \quad \arcsin(x), \quad \arccos(x), \quad \operatorname{arsinh}(x), \ldots$$

[4] https://en.wikipedia.org/wiki/E_(mathematical_constant)

Two points are worth making here. Importantly, for inverses of the above functions to exist, one requires suitable choices for the domain and codomain. For example, we already saw in Example 1.9 that $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ with $f(x) = x^2$ has inverse $f^{-1} : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ with $f^{-1}(y) = \sqrt{y}$ (a power function with power $\alpha = 1/2$), while considered as a mapping $f : \mathbb{R} \to \mathbb{R}$ the function $f(x) = x^2$ is not invertible. Similar considerations apply to logarithmic and inverse trigonometric functions. In general, unless otherwise stated, we will always assume the domain X of any function we consider does not contain any isolated points[5]. Secondly, while the notion of elementary functions may be easy to understand, it is not entirely precise: for example, the hyperbolic functions may be understood as combinations (additions, subtractions, and compositions) of power and exponential functions. The utility of listing the above examples as basic functions will become clear when we consider elementary functions in later lectures, however.

Example 1.17. Check that the inverse function of $\sinh x$ is

$$\operatorname{arsinh}(x) = \ln(x + \sqrt{1 + x^2}).$$

Solution: We substitute to obtain

$$\sinh\left(\ln(x + \sqrt{1 + x^2})\right) = \frac{1}{2}\exp\left(\ln(x + \sqrt{1 + x^2})\right) - \frac{1}{2}\exp\left(-\ln(x + \sqrt{1 + x^2})\right).$$

Since $\ln$ is the inverse of $\exp$ and $\exp(-x) = 1/\exp(x)$, which is well-defined since $\exp(x) > 0$ for all $x \in \mathbb{R}$, we have

$$\sinh\left(\ln(x + \sqrt{1 + x^2})\right) = \frac{1}{2}(x + \sqrt{1 + x^2}) - \frac{1}{2(x + \sqrt{1 + x^2})},$$

which simplifies to

$$\sinh\left(\ln(x + \sqrt{1 + x^2})\right) = x$$

confirming that this is indeed the correct inverse function. □

Usually the functions we will work with in calculus are elementary functions. In fact, using power functions, one can write $|x| = \sqrt{x^2} = (x^2)^{1/2}$, and since x^2 is a power function and $x^{1/2}$ is also a power function on $\mathbb{R}_{\geq 0}$, the modulus is also an elementary function (a composition of power functions). For completeness, we end by giving two examples of non-elementary functions

Example 1.18. The function $\operatorname{sign} : \mathbb{R} \to \mathbb{R}$ given by

$$\operatorname{sign}(x) = \begin{cases} 1 & \text{if } x > 0, \\ 0 & \text{if } x = 0, \\ -1 & \text{if } x < 0 \end{cases} \tag{1.1}$$

is often called the *sign function*. Remark: note that this would be considered an elementary function on $\mathbb{R} \setminus \{0\}$, since then $\operatorname{sign}(x) = |x|/x$, but this is not defined at $x = 0$.

[5] We will not formalise what this means but hopefully it is intuitive.

Example 1.19. The *Dirichlet function* $d : \mathbb{R} \to \mathbb{R}$, defined by the following rule:

$$d(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \notin \mathbb{Q}, \end{cases}$$

for all $x \in \mathbb{R}$. Remark: If studying real analysis, you will see how to work with such "pathological examples". However, in calculus we will usually only consider classes of "well behaved" functions.

Other examples of non-elementary functions may be defined to satisfy certain derivative or integral relations (that we shall see later in the course).

Chapter 2

Sequences, limits and continuity

2.1 Sequences of real numbers

We continue our study of calculus by studying sequences of real numbers, which we will use to illustrate the important concepts of infinity, limits, and convergence. Some of this material will inevitably overlap with a real analysis module, so if you are taking one you may hear some of the same material twice. However, here we will only be introducing the minimal amount of formality, sufficient to get a good grasp of these concepts but not exploring them in their full glory. Hopefully these two approaches are complementary if you are taking a real analysis course.

2.1.1 Concept of a sequence

Formally, a sequence of real numbers is a function $x : \mathbb{N} \to \mathbb{R}$ that maps each $n \in \mathbb{N}$ to a real number $x_n = x(n) \in \mathbb{R}$. Informally, a sequence of real numbers is therefore an ordered list of real numbers $x_1, x_2, x_3, ...$, which goes on forever, also written as (x_n) or $(x_n)_{n=1}^{\infty}$. We have previously discussed the cardinality of the infinite set $\mathbb{N}$, and therefore the formal definition of a sequence of real numbers as a function shows that, since an output must be defined for every point in the domain, we are immediately faced with the concept of infinity. Namely, to fully understand a sequence of real numbers we will need to understand how it behaves as n grows ever larger, i.e. characterise its *asymptotic behaviour*, which will be formally introduced in the next subsections.

For now, we constrain ourselves to a few examples of sequences up to some finite value of $n \in \mathbb{N}$, which allows us to graph out some of their characteristics as in Figure 2.1. Note that, if one wishes to describe a sequence, it is not enough to list just the first few terms. One must describe every term. In these examples there are nice formulae for the nth term of each sequence, and so it suffices to give this formula in each case. However, sometimes sequences are defined rather more implicitly, such as via a recursive formula for example.

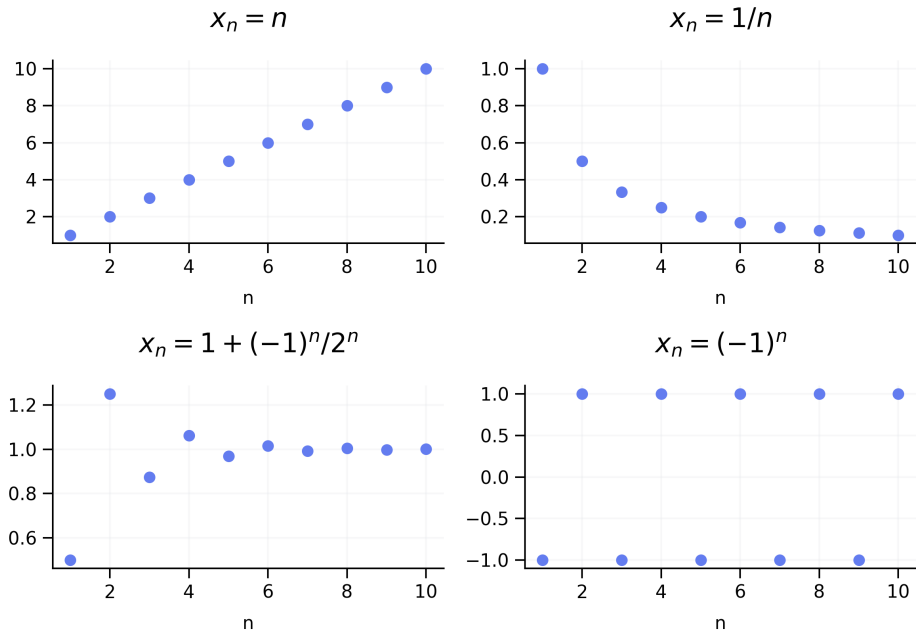


Figure 2.1: Figure illustrating the behaviour of sequences of real numbers corresponding to examples described in text, from $n = 1, 2, ..., 10$. The blue dots correspond to the values of x_n (y-axis) at the corresponding values of n (x-axis).

Example 2.1. The first ten values in the sequence $(x_n)_{n=1}^{\infty}$ given by $x_n = n$ are displayed in the top left panel of Figure 2.1. We see that in this case the values x_n will continue to grow for ever increasing values of n, which means whatever large number we choose, say $K \in \mathbb{R}$, we can always find some $N \in \mathbb{N}$ such that $x_N = N > K$. We will formalise this notion to say what it means for a sequence to be "tending to infinity" in the next subsection.

Example 2.2. The first ten values in sequence $(x_n)_{n=1}^{\infty}$ given by $x_n = 1/n$ are displayed in the top right panel of Figure 2.1. We see that in this case the values x_n will get closer and closer to 0 for ever increasing values of n. We will formalise this notion to say what it means for a sequence to be "converging" (in this case to zero) in a following subsection.

Example 2.3. The first ten values in sequence $(x_n)_{n=1}^{\infty}$ given by $x_n = 1 + (-1)^n/2^n$ are displayed in the bottom left panel of Figure 2.1. We see that in this case the values x_n will get closer and closer to 1 for ever increasing values of n. Similarly to the previous example, we will formalise this notion to say what it means for a sequence to be "converging" (in this case to one) in a following subsection.

Example 2.4. The first ten values in sequence $(x_n)_{n=1}^{\infty}$ given by $x_n = (-1)^n$ are displayed in the bottom right panel of Figure 2.1. We see that in this case the values x_n will continue

to "oscillate" between -1 and $+1$, for ever increasing values of n. Unlike the previous examples, this sequence appears not to be "tending to infinity" but nor does it seem to be "converging" to any particular value. Note that this sequence can also be defined recursively, taking $x_1 = -1$ and then defining $x_n = -x_{n-1}$ for all $n = 2, 3, ...$

2.1.2 Tending to infinity and bounded sequences

When looking at the last three examples of sequences from the previous subsection it is clear that, if we were to continue adding more values of n in Figure 2.1, we would not need to extend the y-axes for their plots. However, for the first example given by $x_n = n$, adding more values of n would mean we must also continue extending the y-axis to include ever greater values of x_n. To capture the way this notion distinguishes some sequences from others, we introduce the following definition.

Definition 2.5. (Tends to infinity). A sequence $(x_n)_{n=1}^{\infty}$ of real numbers *tends to infinity* if given any real number $K > 0$ there exists $N \in \mathbb{N}$ such that

$$x_n > K \quad \text{for all} \quad n > N.$$

We sometimes write "$x_n \to \infty$" or "x_n tends to infinity" instead of "$(x_n)_{n=1}^{\infty}$ tends to infinity".

Example 2.6. The sequence $(x_n)_{n=1}^{\infty}$ given by $x_n = n$ clearly tends to infinity. To show this note that, as described before, for any $K \in \mathbb{R}$ we can always choose $N \in \mathbb{N}$ such that $N > K$ and therefore $x_n > x_N = N > K$ for all $n > N$.

Example 2.7. The sequence $(x_n)_{n=1}^{\infty}$ given by $x_n = \sqrt{n}$ also tends to infinity. To show this, let $K \in \mathbb{R}$. We want to find $N \in \mathbb{N}$ such that $\sqrt{n} > K$ for all $n > N$. Notice that if $n > K^2$ then $\sqrt{n} > K$. Thus, we take $N \geq K^2$, which implies that for all $n > N \geq K^2$ we have $\sqrt{n} > K$, as required.

There is an equivalent definition for a sequence that "tends to minus infinity".

Definition 2.8. (Tends to minus infinity). A sequence $(x_n)_{n=1}^{\infty}$ of real numbers *tends to minus infinity* if given any real number $K > 0$ there exists $N \in \mathbb{N}$ such that

$$x_n < -K \quad \text{for all} \quad n > N.$$

We sometimes write "$x_n \to -\infty$" or "x_n tends to minus infinity" instead of "$(x_n)_{n=1}^{\infty}$ tends to minus infinity". A direct comparison of definitions shows that $x_n \to \infty$ if and only if $-x_n \to -\infty$.

Sequences of real numbers that tend to (plus or minus) infinity are sometimes problematic to work with as they effectively grow forever, but at least the above has given us one concrete definition for the concept of "infinity" as a limiting behaviour of certain sequences. In contrast to sequences tending to infinity is the concept of a bounded sequence of real numbers, which in some sense means the sequence is better behaved.

Definition 2.9. (Bounded sequence). A sequence $(x_n)_{n=1}^{\infty}$ of real numbers is *bounded* if and only if there exists some real number $K \in \mathbb{R}$ such that $|x_n| \leq K$ for all $n \in \mathbb{N}$.

This captures the notion that the other examples of sequences in Figure 2.1 are constrained to lie between some upper and lower values, no matter how large n becomes. There are also notions of bounded from above or below that you may encounter in a real analysis course (a bounded sequence of real numbers is both bounded from above and below). You may ask if tending to infinity or boundedness are the only two possibilities for a sequence of real numbers: unfortunately, they are not!

Example 2.10. The sequence $(x_n)_{n=1}^{\infty}$ given by $x_n = (-1)^n \cdot n$ is neither bounded nor tends to (plus or minus) infinity. Try plotting the sequence yourself to gain intuition for this.

Finally, what distinguishes the different properties of bounded sequences? This involves the notion of convergence, which forms one of the most important attributes of sequences of real numbers and leads naturally to the concept of limits and continuity of real functions.

2.1.3 Convergent sequences

Returning to Figure 2.1, we recall that among the three bounded sequences, two are distinguished by seemingly getting closer and closer to some fixed real number as n increases. Previously, we saw that the modulus $|x-y|$ of the difference of two real numbers $x, y \in \mathbb{R}$ can be interpreted as a distance on $\mathbb{R}$, since it is symmetric, non-negative and satisfies the triangle inequality. Thus, we can partly formalise the notion of "getting close to some $x^* \in \mathbb{R}$" by saying that $|x_n - x^*|$ should become "small" for large values of n. Borrowing from what one typically sees in a real analysis course, we use $\epsilon > 0$ to denote an arbitrarily small but strictly positive real number. The concept of "small" then becomes concrete by saying that $|x_n - x^*| < \epsilon$ (for sufficiently large n) no matter what $\epsilon > 0$ we choose. This motivates the precise definition of a convergent sequence of real numbers.

Definition 2.11. (Convergent sequence). A sequence $(x_n)_{n=1}^{\infty}$ of real numbers *converges to a real number* x^* if given any $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|x_n - x^*| < \epsilon \quad \text{for all} \quad n > N.$$

If a sequence $(x_n)_{n=1}^{\infty}$ converges to some $x^* \in \mathbb{R}$ we say that it *converges* or *tends to* x^*, and write $x_n \to x^*$ or sometimes

$$\lim_{n\to\infty} x_n = x^*.$$

We say that x^* is the *limit* of the convergent sequence $(x_n)_{n=1}^{\infty}$; hence, we now see a concrete definition of a limit in this course. Note again that $\pm\infty$ are not real numbers and so a sequence of real numbers that tends to (plus or minus) infinity is not a convergent sequence. Indeed, it turns out that *every convergent sequence is a bounded sequence* but the converse is not necessarily true.

Example 2.12. To show the sequence $(x_n)_{n=1}^{\infty}$ given by $x_n = 1/n$ converges to zero we need to show that, for any $\epsilon > 0$, we can find an $N \in \mathbb{N}$ such that

$$|x_n - 0| = \left|\frac{1}{n} - 0\right| = \left|\frac{1}{n}\right| = \frac{1}{n} < \epsilon \quad \text{for all} \quad n > N.$$

If we take a natural number $N > 1/\epsilon$ we have $1/N < \epsilon$ and clearly then for any $n > N$ we have $1/n < \epsilon$ as well. Since ϵ was arbitrary we therefore conclude $x_n \to 0$.

Example 2.13. It is straightforward to show that the sequence $(x_n)_{n=1}^{\infty}$ given by $x_n = (-1)^n$ is bounded since for all $n \in \mathbb{N}$ we have $|x_n| = |(-1)^n| = 1$. However, this sequence is not convergent (see Figure 2.1 for intuition and real analysis course for proof).

While beyond the scope of this course, one can prove that the limit of a convergent sequence of real numbers is *unique* and that there is an algebra of limits analogous to the algebra of limits of real functions, which we will introduce in the next section. If you are taking the real analysis course, you will also learn how to establish convergence and calculate limits of sequences of real numbers far more complicated that the examples we have considered here. In these lectures, we have only introduced the bare concepts required to study certain properties of real functions encountered in calculus.

2.2 Limits and continuity of real functions

We now turn to the notions of limits and continuity of real functions (functions whose inputs and outputs are real numbers). You are perhaps already intuitively familiar with these concepts, but it is important to have a concrete grasp of their precise mathematical

meaning in order to avoid intuition leading to the wrong mathematical conclusions, as can sometimes happen, especially when dealing with infinity[1]. While we will only introduce the minimum formality required for calculus, you will see much more like this if you are taking a real analysis course.

2.2.1 Continuity of real functions via convergent sequences

In the previous section we introduced and plotted sequences of real numbers, including some that were convergent, as formally defined in Definition 2.11. Since the codomain of any real sequence $(x_n)_{n=1}^{\infty}$ is a subset of $\mathbb{R}$ (recall we can formally think of a sequence of real numbers as a function $x : \mathbb{N} \to \mathbb{R}$), given a generic real function f we can obtain another real sequence $(f(x_n))_{n=1}^{\infty}$, simply by taking each output $f(x_n)$ as the nth term in the new sequence. Informally, we could write this new sequence as $f(x_1), f(x_2), f(x_3), \ldots$. We should, however, be careful to ensure that the function is well-defined on each term x_n. That is to say, if $f : X \to \mathbb{R}$, then we must restrict ourselves to sequences such that $x_n \in X$ for all $n \in \mathbb{N}$, in which case we say the *sequence* $(x_n)_{n=1}^{\infty}$ *is within the domain* X.

To illustrate this concept, in Figure 2.2 we apply the function $f(x) = x^2$ to the examples from the previous section. Note that in this case we are plotting these new sequences in a way that will help us gain intuition for the definition of continuity in terms of convergent sequences.

Example 2.14. The first ten values obtained after applying the function $f(x) = x^2$ to the sequence with $x_n = n$ from Example 2.1 are displayed in the top left panel of Figure 2.2. We see that the new sequence $(f(x_n))_{n=1}^{\infty}$, given by $f(x_n) = n^2$, will clearly tend to infinity also, and leave it as an exercise for you to prove this. This representation results in the familiar parabolic curve of the function $f(x) = x^2$ that is expected when graphing functions in this way.

Example 2.15. The first ten values obtained after applying the function $f(x) = x^2$ to the sequence with $x_n = 1/n$ from Example 2.2 are displayed in the top right panel of Figure 2.2. In this case we have $f(x_n) = 1/n^2$ and so the sequence starts with the $n = 1$ term on the far right and then moves inwards towards zero as both converge to 0 (exercise to prove this).

Example 2.16. The first ten values obtained after applying the function $f(x) = x^2$ to the sequence with $x_n = 1 + (-1)^n/2^n$ from Example 2.3 are displayed in the bottom left panel of Figure 2.2. The new sequence obtained in this way will also converge to 1.

[1]Here is a "proof" that $0 = 1 : 0 = 1-1+1-1+1-1+\cdots = 1+(-1+1)+(-1+1)+(-1+1)+\cdots = 1+0+0+0+\cdots = 1$. This is clearly nonsense but identifying what went wrong based on intuition alone is difficult. It requires the formal treatment of infinite series to resolve, which you will see if taking a real analysis course.

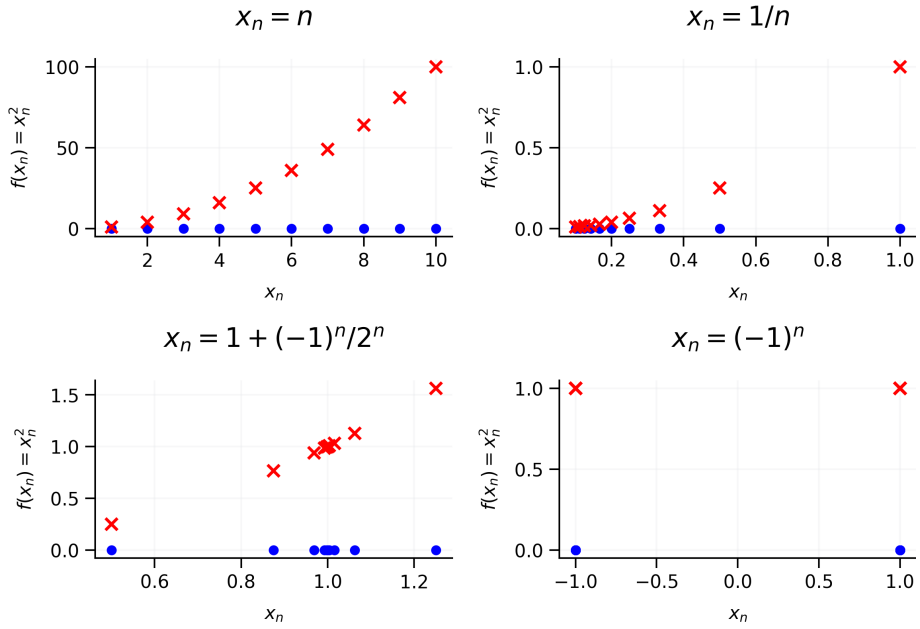


Figure 2.2: Visualisation of the first ten terms in sequences $(x_n)_{n=1}^{\infty}$ and $(f(x_n))_{n=1}^{\infty}$, where the latter is obtained by applying the function $f(x) = x^2$ to the examples described in text. In this representation, the blue dots corresponding to the values x_n are now plotted on the x-axis (real interval) and the corresponding values of $f(x_n)$ are now plotted as red crosses on the y-axis.

Example 2.17. The first ten values obtained after applying the function $f(x) = x^2$ to the sequence with $x_n = (-1)^n$ from Example 2.4 are displayed in the bottom right panel of Figure 2.2. In this case $f(x_n) = ((-1)^n)^2 = 1$ and so the new sequence corresponds to the constant sequence $1, 1, 1, \ldots$.

We now use convergent sequences to make the following definition of continuity of a real function.

Definition 2.18. (Continuity at a point). Let $f : X \to \mathbb{R}$ be a function, where $X \subseteq \mathbb{R}$, and let $x^* \in X$. We say that f is *continuous at* x^* if and only if, for every sequence $(x_n)_{n=1}^{\infty}$ in X that converges to x^*, the corresponding sequence $(f(x_n))_{n=1}^{\infty}$ converges to $f(x^*)$.

It must be immediately highlighted that, as per the definition above, continuity is an attribute of a function at a *specific point*, based on the behaviour of the function around that point. There are different ways to extend the notion of continuity to a set. For the purpose of these lectures, a function $f : X \to \mathbb{R}$ is said to be *continuous on* X if and only if it is continuous at every point $x \in X$.

Intuitively, a function is continuous at the point x^* if a small variation around x^* induces at most a small variation around the value $f(x^*)$. Equivalently, if we move away from x^* by an infinitely small amount, there should be no "jump" (known as a *discontinuity*) in f to a value located more than an infinitely small distance from $f(x^*)$. The definition above makes this notion mathematically precise by including *every* sequence of real numbers that converges to x^*. Convergence (Definition 2.11) means progressive terms x_n and $f(x_n)$ get arbitrarily close to x^* and $f(x^*)$, respectively. So, if the function is continuous at x^*, there should be no special way to approach x^* without also approaching $f(x^*)$ (see left hand panel of Figure 2.3 for illustration).

While simple to state, showing a function is continuous directly from Definition 2.18 is cumbersome and rarely done (we would have to show convergence for all possible sequences). On the other hand, it is often straightforward to confirm a function is not continuous (and therefore *discontinuous*) at a given point.

Example 2.19. The function $f : [-1, 1] \to \mathbb{R}$ given by

$$f(x) = \begin{cases} \dfrac{1}{x}, & \text{if } x > 0, \\ 0, & \text{if } x \leq 0 \end{cases}$$

is not continuous at the point 0. To show this, it suffices to find a particular sequence in $[-1, 1]$ that converges to 0 but such that $(f(x_n))_{n=1}^{\infty}$ does not converge to $f(0) = 0$. We take the sequence $(x_n)_{n=1}^{\infty}$ given by $x_n = 1/n$, which from Example 2.12 we know converges to 0 and clearly we have $1/n \in [-1, 1]$ for all $n \in \mathbb{N}$. We find that $f(x_n) = f(1/n) = n$, and from Example 2.6 we know that this new sequence tends to infinity. Thus, we have found a sequence $(x_n)_{n=1}^{\infty}$ converging to 0 such that $(f(x_n))_{n=1}^{\infty}$ does not converge to $f(0)$, and therefore f is discontinuous at 0 (see right hand panel of Figure 2.3 for illustration).

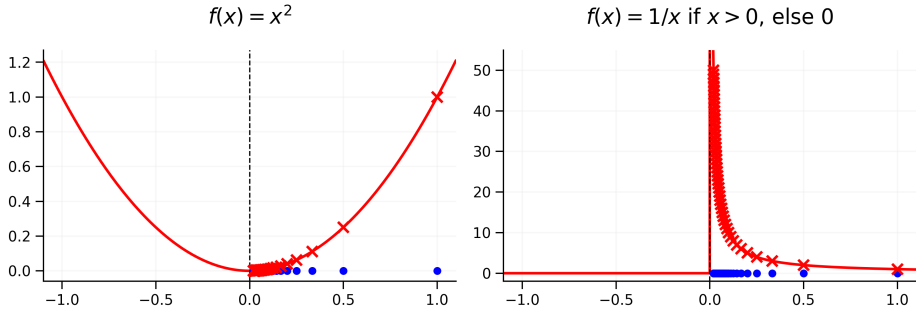


Figure 2.3: Visualisations of continuity and discontinuity defined using convergent sequences. On the left, the function $f : [-1, 1] \to \mathbb{R}$ given by $f(x) = x^2$ is plotted in red, with blue dots representing the sequence given by $x_n = 1/n$ and red crosses denoting corresponding values of $f(x_n)$. Although just one sequence is plotted in this case, it should be clear that any other converging to 0 in $[-1, 1]$ would generate red crosses that similarly converge to $f(0) = 0$, since the function is continuous at 0. On the right, the function $f : [-1, 1] \to \mathbb{R}$ given in Example 2.19 is plotted in red, with blue dots representing the sequence given by $x_n = 1/n$ and red crosses denoting corresponding values of $f(x_n)$. In this case, we see that the red crosses do not converge to $f(0) = 0$ since the function is not continuous at 0.

Why do we care about continuity in calculus? Well, beyond the useful mathematical properties of continuous functions there is a reason going back to the origins of the subject: calculus was developed as a tool for understanding the temporal motion of physical systems. That is, real functions of a real variable $t \in \mathbb{R}$ interpreted as time were used to represent physical quantities, such as how the position of a particle changes in time. Now, if such a function is discontinuous this implies there must be an instantaneous jump in the particle's position at some moment in time (Figure 2.4). Such a phenomenon would represent teleportation, certainly not thought to be physically possible! While I cannot guarantee that all functions we will encounter in this course will be continuous, we can state a final, highly important result (in bold for emphasis):

Result 2.20. All elementary functions are continuous on their natural domains.

This result can be obtained by showing the basic functions in Section 1.2.4 are continuous on their natural domains, and then that combinations of them are also continuous. You will do this if you are taking a real analysis course.

2.2.2 Limits of functions

In the previous subsection, we used limits of converging sequences to define what it means for a function to be continuous at a point in its domain. Informally, Definition 2.18 says

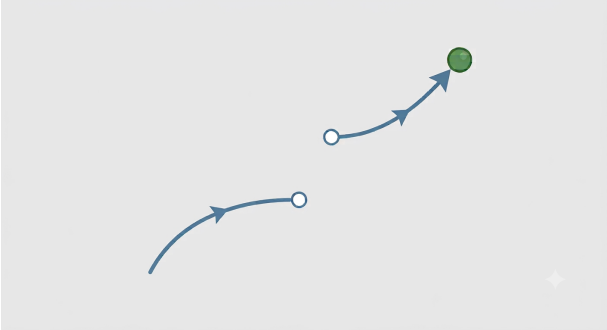


Figure 2.4: Cartoon schematic of a particle (green) moving along a discontinuous trajectory (blue) with discontinuity represented by open circles.

that $f(x)$ approaches $f(x^*)$ as x approaches x^*. In this subsection we introduce the notation

$$\lim_{x \to x^*} f(x)$$

for the value that $f(x)$ approaches as x approaches some $x^* \in \mathbb{R}$, and we will see that, under a mild condition, f is continuous at x^* if and only if

$$\lim_{x \to x^*} f(x) = f(x^*).$$

You may have seen this type of notation before, involving *limits of functions*.

Importantly, we can use limits of functions to study the behaviour of f independent from the value $f(x^*)$, for example even if f is not continuous at x^* or x^* is not contained in the domain of f. More generally, the limit $L \in \mathbb{R}$ (if it exists) of a function $f : X \to \mathbb{R}$ as $x \to x^*$ is denoted

$$\lim_{x \to x^*} f(x) = L$$

and is defined if and only if:

1. There exists at least one sequence $(x_n)_{n=1}^{\infty}$ in $X \setminus \{x^*\}$ such that $x_n \to x^*$; and
2. For every sequence $(x_n)_{n=1}^{\infty}$ in $X \setminus \{x^*\}$ such that $x_n \to x^*$ we have that $f(x_n) \to L$.

This notation captures the behaviour of $f(x)$ as x approaches the point x^* through points of the domain X excluding the point x^* itself. The same limiting value must be obtained regardless of how x approaches x^* within $X \setminus \{x^*\}$.

As mentioned above, it turns out that an alternative definition of continuity is that a function $f : X \to \mathbb{R}$ is continuous at $x^* \in X$ for which condition 1 holds if and only if the limit of f as $x \to x^*$ exists in the sense defined above and equals $f(x^*)$. On the other hand, x^* may be outside the domain of f or the function may be discontinuous at x^*, but the limit may still exist because it does not depend on the value of f at x^*. Or, the limit may not exist at all.

Example 2.21. The function $f : [-1, 1] \to \mathbb{R}$ given by

$$f(x) = \begin{cases} x, & \text{if } x \neq 0, \\ 1, & \text{if } x = 0 \end{cases}$$

is not continuous at 0 (exercise to show this using Definition 2.18). Intuitively, we see the function $f(x)$ will get arbitrarily close to 0 as x approaches 0 (Figure 2.5a), so we have

$$\lim_{x \to 0} f(x) = 0.$$

Rather than proving this directly from the definition of the limit, we note that on $[-1, 1] \setminus \{0\}$ the function given by $g(x) = x$ agrees with $f(x)$ but g is also well-defined as an elementary (and therefore continuous) function on the whole of $[-1, 1]$. Moreover, since the limit L does not depend on the value of $f(0)$ we can use continuity of g and substitution to obtain

$$\lim_{x \to 0} f(x) = \lim_{x \to 0} g(x) = \lim_{x \to 0} x = 0.$$

Thus, $L = 0 \neq f(0) = 1$, again showing that f is discontinuous at 0.

Example 2.22. The function $f : [-1, 1] \to \mathbb{R}$ given by

$$f(x) = \begin{cases} 1, & \text{if } x > 0, \\ -1, & \text{if } x \leq 0 \end{cases}$$

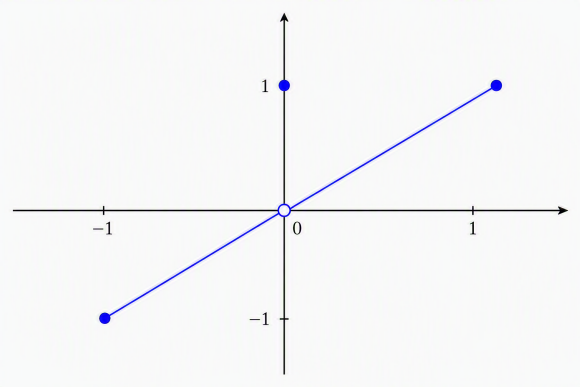
is not continuous at 0 (exercise to show this using Definition 2.18). Intuitively, we see the function $f(x)$ will get arbitrarily close to -1 as x approaches 0 from the left hand side, but will get arbitrarily close to 1 as x approaches 0 from the right hand side (Figure 2.5b). This is typically denoted as

$$\lim_{x \to 0^-} f(x) = -1, \quad \lim_{x \to 0^+} f(x) = 1.$$

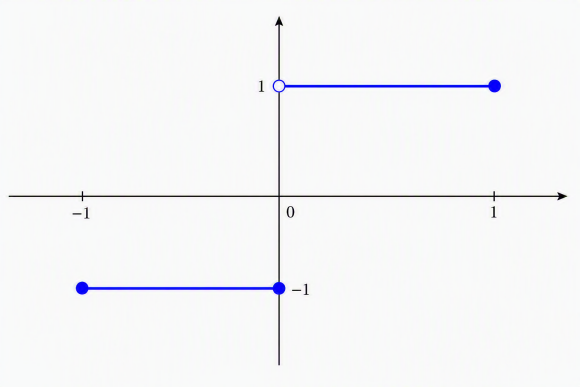
So, the limiting behaviour of the function does not match on either side of 0, and we therefore say the limit does not exist. One-sided limits will not be formalised here, but you will do so if you are taking a real analysis course.

The previous examples were rather contrived to illustrate the concepts of limits of discontinuous functions, which occur infrequently in standard calculus applications. However, there are several relevant extensions of limits that are also required to understand the behaviour of continuous functions. These extensions have definitions involving sequences that you may see if you are taking a real analysis course. In the first case, it may be that the function $f : X \to \mathbb{R}$ *tends to (plus or minus) infinity* as x tends to x^*, which we denote

$$\lim_{x \to x^*} f(x) = \pm\infty.$$



(a) Plot of the function from Example 2.21.



(b) Plot of the function from Example 2.22.

Figure 2.5: Plots of the functions from examples in the main text.

Example 2.23. The function $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ given by $f(x) = 1/x^2$ tends to infinity as x tends to 0, which we denote

$$\lim_{x \to 0} \frac{1}{x^2} = \infty.$$

This example of a limit appears often in calculus applied to physical systems where it is known as the inverse square law. It can represent, for example, the function describing the attractive force of gravity between two particles and becomes infinitely large as the particles get arbitrarily close together.

Actually, the previous example involves a slight abuse of notation because it seemingly implies that the limit exists with $L = \pm\infty$; however, since $L \in \mathbb{R}$ this cannot be true, and so even though it is written this way we still say the limit does not exist. Note that in the example above the point x^* is not actually in the domain of f, and so does not contradict the fact that the function is continuous on X.

In the second case, we may be interested in the behaviour of the function $f : X \to \mathbb{R}$ as x itself tends to (plus or minus) infinity. We denote this as

$$\lim_{x \to \pm\infty} f(x) = L,$$

where again by abuse of notation we use L to denote either a real number, if this limit exists, or $\pm\infty$ if f also tends to (plus or minus) infinity in this case.

Example 2.24. The function $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ given by $f(x) = 1/x$ tends to 0 as x tends to infinity, which we denote

$$\lim_{x \to \infty} \frac{1}{x} = 0.$$

On the other hand, the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x$ tends to infinity as x tends to infinity, which we denote as

$$\lim_{x \to \infty} x = \infty.$$

While calculating limits of functions directly from the definition can be difficult, in calculus we will be helped by dealing almost exclusively with elementary functions. Since these are continuous, many limits can usually be calculated by direct substitution if involving points within their natural domain as we did in Example 2.21. However, this may not always be the case or their natural domain may not always be obvious. Thus, for completeness we introduce, without proof, the algebra of limits in the form of the following result.

Result 2.25. (Algebra of limits). Let $f, g : X \to \mathbb{R}$. Suppose

$$\lim_{x \to x^*} f(x) = L_f \quad \text{and} \quad \lim_{x \to x^*} g(x) = L_g$$

exist and $c \in \mathbb{R}$ is a constant. Then:

1. **(Constants)**
$$\lim_{x \to x^*} c \cdot f(x) = c \cdot L_f;$$

2. **(Sums)**
$$\lim_{x \to x^*} (f(x) \pm g(x)) = L_f \pm L_g;$$

3. **(Products)**
$$\lim_{x \to x^*} f(x)g(x) = L_f L_g;$$

4. **(Quotients)**
$$\lim_{x \to x^*} \frac{f(x)}{g(x)} = L_f / L_g, \quad \text{provided} \quad L_g \neq 0;$$

5. **(Powers)**
$$\lim_{x \to x^*} f(x)^{g(x)} = L_f^{L_g}, \quad \text{provided} \quad L_f > 0.$$

These rules are also valid for x tending to $\pm\infty$. However, the algebra of limits gives no information when the behaviour of $f(x)$ and $g(x)$ combines into one of the following, known as *limits of indeterminate form*:

$$\frac{0}{0}, \quad \frac{\pm\infty}{\pm\infty}, \quad 0 \cdot (\pm\infty), \quad \infty - \infty, \quad 0^0, \quad \infty^0, \quad 1^{\pm\infty}.$$

In these cases the limit may or may not exist, and must be found by other means. We will see how to effectively deal with some limits of indeterminate form in the examples below, and some alternative methods will arise later in the course.

Example 2.26. Calculate the limit

$$\lim_{x\to 1} \frac{x^2-1}{x-1}.$$

Solution: Direct substitution at $x^* = 1$ yields the indeterminate form $0/0$, since both the numerator and denominator of the rational function $f(x) = (x^2-1)/(x-1)$ vanish at $x = 1$. As a result, $x^* = 1$ is outside the natural domain of f, so continuity cannot be applied directly to evaluate the limit at this point.

However, $f(x)$ is an elementary function since it is constructed as the quotient of two polynomials. For all $x \neq 1$, we factorise the numerator to simplify the expression as

$$f(x) = \frac{(x-1)(x+1)}{x-1} = x+1 \quad \text{for } x \neq 1.$$

Because the limit of $f(x)$ as $x \to 1$ depends only on the values of $f(x)$ for x close to 1 (and never at $x = 1$ itself), we can replace $f(x)$ with the simplified function $x+1$, which is an elementary function continuous everywhere on $\mathbb{R}$. Thus, we can now evaluate the limit using direct substitution

$$\lim_{x\to 1} \frac{x^2-1}{x-1} = \lim_{x\to 1}(x+1) = 1+1 = 2.$$

□

Example 2.27. Calculate the limit

$$\lim_{x\to\infty} \frac{3x^2+2x-1}{x^2-2}.$$

Solution: Although the function $f(x) = (3x^2+2x-1)/(x^2-2)$ is undefined for $x = \pm\sqrt{2}$ this doesn't affect the limit as $x \to \infty$ and $f(x)$ is an elementary function since it is constructed as the quotient of two polynomials. However, direct application of the algebra of limits from Result 2.25 to the function in its current form would result in an indeterminate form ∞/∞.

Instead, for $x > \sqrt{2}$, we can rearrange the expression to give

$$f(x) = \frac{3x^2+2x-1}{x^2-2} = \frac{3+\frac{2}{x}-\frac{1}{x^2}}{1-\frac{2}{x^2}} \quad \text{for } x > \sqrt{2}$$

and in this case by the algebra of limits we have[2]

$$\lim_{x\to\infty} \frac{3x^2+2x-1}{x^2-2} = \lim_{x\to\infty} \frac{3+\frac{2}{x}-\frac{1}{x^2}}{1-\frac{2}{x^2}} = \frac{\lim_{x\to\infty}\left(3+\frac{2}{x}-\frac{1}{x^2}\right)}{\lim_{x\to\infty}\left(1-\frac{2}{x^2}\right)} = \frac{3}{1} = 3.$$

□

[2]In this solution, we make use of the limit $\lim_{x\to\infty} \frac{1}{x} = 0$, which is taken from Example 2.24.

Example 2.28. Calculate the limit

$$\lim_{x\to 0} \frac{\sqrt{x+4}-2}{x}.$$

Solution: Direct substitution at $x^* = 0$ yields the indeterminate form $0/0$. Because the numerator of $f(x) = (\sqrt{x+4}-2)/x$ contains a square root, factoring polynomials directly as we did in previous examples is not possible.

However, for $x \neq 0$ and $x \geq -4$, we can rationalise the numerator by multiplying top and bottom by $\sqrt{x+4}+2$ to give

$$f(x) = \frac{\sqrt{x+4}-2}{x} \cdot \frac{\sqrt{x+4}+2}{\sqrt{x+4}+2} = \frac{x+4-4}{x(\sqrt{x+4}+2)} = \frac{1}{\sqrt{x+4}+2} \quad \text{for } x \neq 0, x \geq -4.$$

Since the limit as $x \to 0$ does not depend on the values of $f(x)$ at $x = 0$, and $1/(\sqrt{x+4}+2)$ is continuous at $x = 0$, we can evaluate by direct substitution to give

$$\lim_{x\to 0} \frac{\sqrt{x+4}-2}{x} = \lim_{x\to 0} \frac{1}{\sqrt{x+4}+2} = \frac{1}{\sqrt{0+4}+2} = \frac{1}{4}.$$

□

We also state (without proof) a result on limits involving compositions of real functions.

Result 2.29. (Limits of compositions). Let $f : X \to \mathbb{R}$ and $g : Y \to \mathbb{R}$ be functions such that $f(x) \in Y$ for all $x \in X$. Suppose

$$\lim_{x\to x^*} f(x) = L_f$$

and g is continuous at L_f. Then

$$\lim_{x\to x^*} g(f(x)) = g(L_f).$$

This result also holds when taking limits at infinity ($x^* = \pm\infty$), provided $\lim_{x\to\pm\infty} f(x) = L_f$ is a finite real number where g is continuous. If $L_f = \pm\infty$ then

$$\lim_{x\to x^*} g(f(x)) = \lim_{y\to\pm\infty} g(y),$$

provided the limit on the right hand side exists or is $\pm\infty$.

Example 2.30. Calculate the limit

$$\lim_{x\to 1} \cos\left(\frac{\pi(x^2-1)}{3(x-1)}\right).$$

Solution: Let the inner function be $f(x) = \pi(x^2-1)/(3(x-1))$ and the outer function be $g(y) = \cos(y)$. For $x \neq 1$, we simplify the inner algebraic expression by factorisation

$$f(x) = \frac{\pi(x-1)(x+1)}{3(x-1)} = \frac{\pi(x+1)}{3} \quad \text{for } x \neq 1.$$

Taking the limit of the inner function then yields

$$L_f = \lim_{x\to 1} f(x) = \lim_{x\to 1} \frac{\pi(x+1)}{3} = \frac{2\pi}{3},$$

and because $g(y) = \cos(y)$ is continuous everywhere on $\mathbb{R}$ (and in particular at $y = 2\pi/3$), Result 2.29 allows us to evaluate the limit directly

$$\lim_{x\to 1} \cos\left(\frac{\pi(x^2-1)}{3(x-1)}\right) = \cos\left(\lim_{x\to 1} \frac{\pi(x^2-1)}{3(x-1)}\right) = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}.$$

□

Example 2.31. From Example 2.23, we have

$$\lim_{x\to 0} \frac{1}{x^2} = +\infty.$$

Here we also state, without proof, that

$$\lim_{y\to+\infty} \arctan(y) = \frac{\pi}{2}.$$

Thus, from Result 2.29 we have

$$\lim_{x\to 0} \arctan\left(\frac{1}{x^2}\right) = \frac{\pi}{2}.$$

Finally, we end with a short list of notable limits that will be useful later in the course and for your problem sheet exercises:

$$\lim_{x\to 0} \frac{\sin(x)}{x} = 1; \tag{2.1}$$

$$\lim_{x\to 0} \frac{\cos(x)-1}{x} = 0; \tag{2.2}$$

$$\lim_{x\to 0} (1+x)^{1/x} = e; \tag{2.3}$$

$$\lim_{x\to 0} \frac{e^x - 1}{x} = 1; \tag{2.4}$$

$$\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1; \tag{2.5}$$

$$\lim_{x\to \pm\infty} \left(1 + \frac{1}{x}\right)^x = e. \tag{2.6}$$

Again, why are limits of functions important in calculus? As we shall see in the next lectures, the mathematical notions of differentiation and integration are ultimately formulated in terms of infinitesimally small quantities, and thus the entire basis for the topic is grounded in the theory of limits!

Chapter 3

Differentiation

3.1 Definition of the derivative

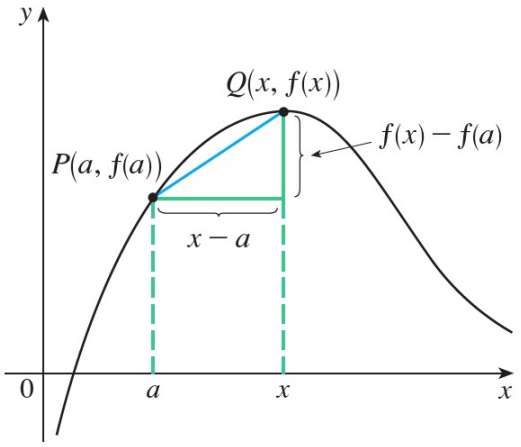
You have probably come across the derivative (of a real function) in some context before. It is central to calculus, since it describes the instantaneous rate of change of a function with respect to its input variable. The process of finding a derivative is called differentiation. We will begin by motivating a formal definition of the derivative in terms of a limit, and then provide derivatives of certain basic functions.

3.1.1 Tangent lines and velocities

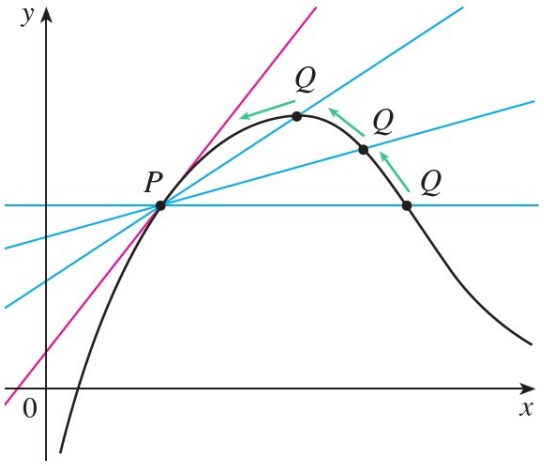
The derivative of a real function of a single real variable at a chosen point, when it exists, is the slope of the tangent line to the graph of the function at that point. The tangent line is the best linear approximation of the function near that input value. Consider a curve (or graph) C that has equation $y = f(x)$ for some function $f : X \to \mathbb{R}$ (by this we mean the set of all ordered pairs $C = \{(x, f(x)) : x \in X\} \subseteq \mathbb{R} \times \mathbb{R}$). Suppose that we want to find an approximation for the slope of the tangent line to C at the point $P = (a, f(a))$ for some $a \in X$. Then we can consider a nearby point $Q = (x, f(x))$, where $x \in X, x \neq a$, and compute the slope of the line PQ (see Figure 3.1a) by taking the ratio of the difference $f(x) - f(a)$ to the difference $x - a$ as follows:

$$\frac{f(x) - f(a)}{x - a}.$$

Intuitively, the further Q is from P, the less accurate this approximation to the slope of the tangent line may be. However, we can let $Q \to P$ (Q approach P) along the curve C by letting $x \to a$ in the above expression. Then, as x gets arbitrarily close to a we hope that this approximation will get arbitrarily close to the true value of the slope (see Figure



(a) Illustration of curve given by $y = f(x)$ with calculation of the slope of the line PQ.



(b) Schematic of line PQ approximating the tangent line (red) as $x \to a$.

3.1b). This motivates defining the slope of the tangent line to the curve $y = f(x)$ at the point $P = (a, f(a))$, when it exists, to be

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a}. \tag{3.1}$$

It is often useful to re-write (3.1) after making a change of variables $h = x - a$ as

$$\lim_{h \to 0} \frac{f(a + h) - f(a)}{h}.$$

To see an example of this relevant to the origins of calculus as the study of real world physics, we could imagine C to be the curve traced out as a particle moves according to an equation of motion of the form $s = f(t)$, where $f(t)$ gives the position of the particle at some time t. Over the time interval $[a, a+h]$, the displacement or change in position of the particle is $f(a+h) - f(a)$, and the average velocity over this time interval is therefore given by

$$\text{average velocity} = \frac{\text{change in position}}{\text{change in time}} = \frac{f(a + h) - f(a)}{h}.$$

Now suppose we compute the average velocities over shorter and shorter time intervals $[a, a+h]$, i.e. we let h approach 0. We define the velocity (or instantaneous velocity) $v(a)$ at time $t = a$ to be the limit of these average velocities

$$v(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}.$$

Hence, limits of this form arise whenever we calculate a rate of change in any of the sciences or engineering, such as a rate of a reaction in chemistry or a marginal cost in economics. Since this type of limit occurs so widely, it is given a special name and notation.

3.1.2 Differentiability and the derivative

The special name and notation referenced to above is for the derivative. For simplicity, in the first instance we will restrict our definition to functions of the form $f : (a, b) \to \mathbb{R}$ where either $a, b \in \mathbb{R}$ with $b > a$ or $a = -\infty$ or $b = \infty$, i.e. whose domains are open intervals[1].

Definition 3.1. A real function $f : (a, b) \to \mathbb{R}$ is *differentiable* at a point $x_0 \in (a, b)$ if the limit

$$\lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} \tag{3.2}$$

exists. If this limit exists, it is called the *derivative* of f at x_0, and is denoted by $f'(x_0)$.

The limit in (3.2) is equivalent to

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

Common alternative notation for $f'(x_0)$ is

$$\frac{\mathrm{d}f}{\mathrm{d}x}(x_0) \quad \text{or, if we write } y = f(x), \qquad \left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=x_0}.$$

In Definition 3.1, we have been careful to specify "if the limit exists". If the limit does not exist, then naturally we say that f is not differentiable at the point x_0. The following is an important result that says differentiability at a point implies continuity at the point.

Result 3.2. If a real function $f : (a, b) \to \mathbb{R}$ is differentiable at $x_0 \in (a, b)$, then f is continuous at x_0. Thus, if a real function f is not continuous at x_0, then f is not differentiable at x_0.

The converse of Result 3.2 is not true in general: a function may be continuous at a point without being differentiable there.

Example 3.3.

Consider $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = |x|$ for all $x \in \mathbb{R}$. We have previously seen f is an elementary function, and so this means it is continuous at $x_0 = 0$. However, using definition 3.1 we have that

$$\lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \to 0} \frac{|0 + h| - |0|}{h} = \lim_{h \to 0} \frac{|h|}{h}.$$

Now, notice that for $h \neq 0$ we have that $|h|/h = \operatorname{sign}(h)$ (see Example 1.18). Thus, $|h|/h$ will remain at -1 as h approaches 0 from the left hand side, but will remain at 1 as h approaches 0 from the right. So the limit will not exist, meaning $f(x) = |x|$ is continuous but not differentiable at $x = 0$.

[1] A little more care is required for general domains, which we will touch on below.

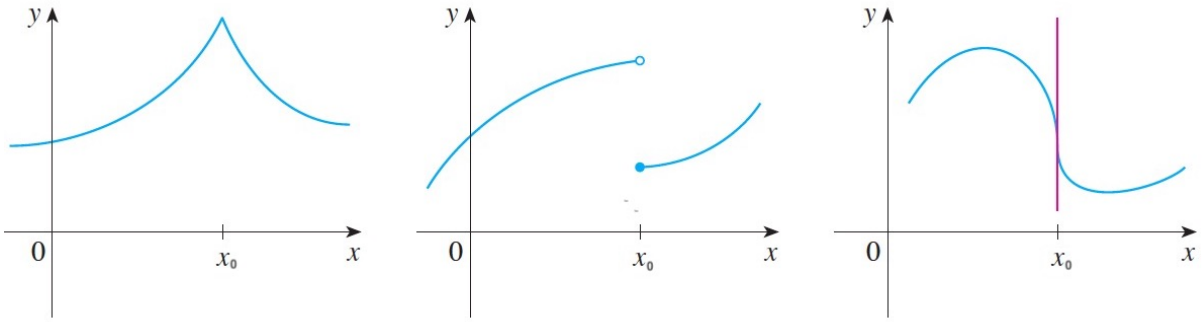


Figure 3.2: Illustrations of ways functions are not differentiable at x_0. Left to right: a corner; a discontinuity; a vertical tangent.

Figure 3.2 depicts three common ways (there are others!) that a function can fail to be differentiable at a point x_0.

While Definition 3.1 is only stated for functions defined on some open interval, the one-sided limit notation extends differentiability naturally to domain boundaries, such as endpoints of closed intervals $[a, b]$. If this is done properly, then Definition 3.1 extends to define the derivative of $f : X \to \mathbb{R}$ at any point of the domain X. We can then consider f' as a new function $f' : Y \to \mathbb{R}$ defined by $f'(y)$ for all $y \in Y$ where the domain $Y \subseteq X$ is the subset of X where the derivative exists. In particular, we call a function $f : X \to \mathbb{R}$ *differentiable* if it is differentiable at every point $x \in X$. Higher-order derivatives are the result of differentiating a function repeatedly. Given that f is a differentiable function, the derivative of f', if it exists, is the second derivative, denoted as f''. The derivative of f'', if it exists, is the third derivative, denoted as f'''. By continuing this process the nth derivative, if it exists, is the derivative of the $(n-1)$th derivative or the derivative of order n. This generalisation of the derivative of a function f may be denoted as $f^{(n)}$. Equivalently, we can use notation

$$\frac{\mathrm{d}f}{\mathrm{d}x}, \quad \frac{\mathrm{d}^2 f}{\mathrm{d}x^2}, \quad \ldots, \quad \frac{\mathrm{d}^n f}{\mathrm{d}x^n}$$

for the derivatives of f up to order n.

3.1.3 Derivatives of some real functions

We now run through a selection of examples directly computing the derivatives of the basic functions (introduced in Subsection 1.2.4) directly from Definition 3.1. You have probably seen and used these results before, and now you will understand where they come from. Importantly, since elementary functions are combinations of these basic functions, we will then be able to compute derivatives of elementary functions using rules described in the next section.

Example 3.4. If $f(x) = c$ with $c \in \mathbb{R}$ for all $x \in \mathbb{R}$ is the constant function, then

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{c - c}{h} = 0$$

and so the derivative of the constant function is identically zero.

Example 3.5. If $f(x) = cx$ with $c \in \mathbb{R}$ for all $x \in \mathbb{R}$ then

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{cx + ch - cx}{h} = \lim_{h\to 0} \frac{ch}{h} = \lim_{h\to 0} c = c,$$

so the derivative of a linear function is constant.

Example 3.6. If $f(x) = cx^2$ with $c \in \mathbb{R}$ for all $x \in \mathbb{R}$ then

$$f'(x) = \lim_{h\to 0} \frac{c(x+h)^2 - cx^2}{h} = \lim_{h\to 0} \frac{c(x^2 + 2xh + h^2) - cx^2}{h} = \lim_{h\to 0} \frac{c(2xh + h^2)}{h},$$

and after cancelling the factor of h and applying the algebra of limits we have

$$f'(x) = \lim_{h\to 0} c\,(2x + h) = \lim_{h\to 0} c2x + \lim_{h\to 0} ch = c2x + 0 = c2x,$$

showing that the derivative of a quadratic function is linear.

The examples above can be generalised to monomial functions of degree $n \in \mathbb{N}$ using the Binomial Theorem[2] to expand the term $(x+h)^n$ that will appear in the numerator inside the limit, and obtain the following result.

Result 3.7. If $f(x) = cx^n$ with $n \in \mathbb{N}$ and $c \in \mathbb{R}$ for all $x \in \mathbb{R}$, then $f'(x) = cnx^{n-1}$.

In fact, although the rules for derivatives of sums of functions will not be stated until the next section, it should already be clear that using this result combined with the algebra of limits shows that the derivative of any polynomial of degree $n \in \mathbb{N}$ of the form

$$f(x) = \sum_{k=0}^{n} c_k x^k, \quad c_k \in \mathbb{R} \quad \text{for all } k$$

is given by

$$f'(x) = \sum_{k=1}^{n} c_k k x^{k-1}.$$

For general power functions we have the following result:

[2]Do not worry if you have not seen this before, it is only about knowing the same concept applies!

Result 3.8. If $f(x) = cx^\alpha$ with $\alpha \in \mathbb{R} \setminus \{0\}$ for all x where this expression defines a real-valued differentiable function, then $f'(x) = c\alpha x^{\alpha-1}$.

We can also use some of the notable limits from Subsection 2.2.2 to compute the derivatives of the basic exponential and trigonometric functions.

Example 3.9. Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = e^{\alpha x}$ for some $\alpha \in \mathbb{R} \setminus \{0\}$. Then,

$$f'(x) = \lim_{h\to 0} \frac{e^{\alpha(x+h)} - e^{\alpha x}}{h} = \lim_{h\to 0} e^{\alpha x}\left[\frac{e^{\alpha h} - 1}{h}\right] = \alpha e^{\alpha x} \lim_{y\to 0} \frac{e^y - 1}{y},$$

where we have substituted $y = \alpha h$ and brought the constant factors outside of the limit. We notice that the notable limit

$$\lim_{y\to 0} \frac{e^y - 1}{y} = 1$$

from Subsection 2.2.2 now gives

$$f'(x) = \alpha e^{\alpha x}.$$

The derivatives of more general exponential functions will be obtained in the next section.

Example 3.10. Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = \sin(\alpha x)$ for some $\alpha \in \mathbb{R} \setminus \{0\}$. Then, using the trigonometric identity

$$\sin(x + y) = \sin(x)\cos(y) + \sin(y)\cos(x)$$

we have

$$f'(x) = \lim_{h\to 0} \frac{\sin(\alpha(x + h)) - \sin(\alpha x)}{h} = \lim_{h\to 0} \frac{\sin(\alpha x)\cos(\alpha h) + \sin(\alpha h)\cos(\alpha x) - \sin(\alpha x)}{h}.$$

We can split the expression inside the limit into two parts, and evaluate either individually, assuming both limits exist so that the algebra of limits applies. Indeed, we have

$$\frac{\sin(\alpha x)\cos(\alpha h) + \sin(\alpha h)\cos(\alpha x) - \sin(\alpha x)}{h} = \sin(\alpha x)\frac{\cos(\alpha h) - 1}{h} + \cos(\alpha x)\frac{\sin(\alpha h)}{h}$$

so after making the substitution $y = \alpha h$ and bringing constant factors outside of the limit, we obtain a combination of two notable limits from Subsection 2.2.2:

$$\begin{aligned} f'(x) &= \alpha \sin(\alpha x) \lim_{y\to 0} \frac{\cos(y) - 1}{y} + \alpha \cos(\alpha x) \lim_{y\to 0} \frac{\sin(y)}{y} \\ &= \alpha \sin(\alpha x) \cdot 0 + \alpha \cos(\alpha x) \cdot 1 = \alpha \cos(\alpha x). \end{aligned}$$

Thus, the derivative of $\sin(\alpha x)$ is $\alpha \cos(\alpha x)$.

It is left as an exercise to show that the derivative of $\cos(\alpha x)$ is $-\alpha\sin(\alpha x)$ by using the trigonometric identity $\cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$. Calculating derivatives of other trigonometric functions will also be shown in the next section when we state the general rules of differentiation. Similarly, we shall combine the rules with the result from Example 3.9 to calculate the derivatives of the hyperbolic functions (you may already guess how this can be done).

The final example presented here involves using another notable limit to calculate the derivative of one very special inverse function, which is the natural logarithm.

Example 3.11. Let $f : (0,\infty) \to \mathbb{R}$ be given by $f(x) = \ln(x)$ for all $x \in (0,\infty)$. Then, since $x > 0$,

$$f'(x) = \lim_{h\to 0} \frac{\ln(x+h) - \ln(x)}{h} = \lim_{h\to 0} \frac{\ln((x+h)/x)}{h} = \lim_{h\to 0} \frac{\ln(1+h/x)}{h}$$

so after making the substitution $y = h/x$ we have

$$f'(x) = \frac{1}{x}\lim_{y\to 0} \frac{\ln(1+y)}{y} = \frac{1}{x},$$

where we used the notable limit

$$\lim_{y\to 0} \frac{\ln(1+y)}{y} = 1.$$

In fact, since $f : (0,\infty) \to \mathbb{R}$ given by $f(x) = \log_a(x)$ with $a > 0, a \neq 1$ can be expressed as $f(x) = \ln(x)/\ln(a)$, we find that more generally that

$$f'(x) = \frac{1}{x\ln(a)} \text{ for all } x \in (0,\infty).$$

3.2 Rules of differentiation

When calculating derivatives in the previous section, we sometimes implicitly used rules derived from the algebra of limits (e.g. bringing a constant term in front of a limit or splitting limit of sums into a sum of limits). Here, we state a central result that collects the rules for differentiation into one framework. Proving these rules will be done if you are taking a real analysis course.

Result 3.12. (Rules of differentiation). Let $\lambda, \mu \in \mathbb{R}$, and let $f, g : (a,b) \to \mathbb{R}$ be differentiable at $x \in (a,b)$.

1. **(Linearity rule)** The linear combination $\lambda f + \mu g$ is differentiable at x, with
$$(\lambda f + \mu g)'(x) = \lambda f'(x) + \mu g'(x).$$
2. **(Product rule)** The product fg is differentiable at x, with
$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$
3. **(Quotient rule)** If $g(x) \neq 0$, the quotient f/g is differentiable at x, with
$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}.$$
4. **(Chain rule)** If $h : (c, d) \to (a, b)$ is differentiable at $y \in (c, d)$ and f is differentiable at $h(y) \in (a, b)$, then the composite function $(f \circ h) : (c, d) \to \mathbb{R}$ is differentiable at y, with
$$(f \circ h)'(y) = f'(h(y))h'(y).$$

We have already implicitly used the linearity rule when stating the result for polynomials (sums of monomials) in the previous results section. Here, we use it to obtain the derivatives of the hyperbolic functions $\cosh(x)$ and $\sinh(x)$.

Example 3.13. Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = \cosh(\alpha x)$ for some $\alpha \in \mathbb{R} \setminus \{0\}$. Then, using the definition in terms of exponential functions and applying the linearity rule we have
$$f'(x) = \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{e^{\alpha x} + e^{-\alpha x}}{2}\right] = \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}x}e^{\alpha x} + \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}x}e^{-\alpha x} = \frac{1}{2}\alpha e^{\alpha x} - \frac{1}{2}\alpha e^{-\alpha x} = \alpha \sinh(\alpha x)$$
where the intermediate result is obtained from Example 3.9. It is straightforward to apply the same approach to show that the derivative of $\sinh(\alpha x)$ is $\alpha \cosh(\alpha x)$ (note the sign– do not get confused with trigonometric functions!).

The product rule can be used to verify Result 3.7 using proof by induction.

Example 3.14. We want to show that for $n \in \mathbb{N}$ and $c \in \mathbb{R}$, we have $d(cx^n)/\mathrm{d}x = cnx^{n-1}$. We proceed by induction on n. From Example 3.5 we have the base case ($n = 1$):
$$\frac{\mathrm{d}}{\mathrm{d}x}(cx^1) = c = c \cdot 1 \cdot x^0.$$
For the inductive step, assume $d(cx^k)/\mathrm{d}x = ckx^{k-1}$ for some $k \geq 1$. For $n = k + 1$, applying the product rule yields:
$$\frac{\mathrm{d}}{\mathrm{d}x}(cx^{k+1}) = c\frac{\mathrm{d}}{\mathrm{d}x}(x^k x)$$

$$
\begin{aligned}
&= c\left[x\frac{\mathrm{d}}{\mathrm{d}x}(x^k) + x^k\frac{\mathrm{d}}{\mathrm{d}x}(x)\right] \\
&= c\left[(kxx^{k-1}) + x^k(1)\right] \\
&= c\left[kx^k + x^k\right] \\
&= c(k+1)x^k.
\end{aligned}
$$

By mathematical induction, the result holds for all $n \in \mathbb{N}$.

The quotient rule can be applied to calculate the derivatives of other trigonometric and hyperbolic functions.

Example 3.15. Let f be given by $f(x) = \sin(\alpha x)/\cos(\alpha x) = \tan(\alpha x)$ on its domain of definition for some $\alpha \in \mathbb{R} \setminus \{0\}$. By the quotient rule,

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\tan(\alpha x) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\sin(\alpha x)}{\cos(\alpha x)}\right) = \frac{1}{(\cos(\alpha x))^2}\left[\cos(\alpha x)\frac{\mathrm{d}}{\mathrm{d}x}\sin(\alpha x) - \sin(\alpha x)\frac{\mathrm{d}}{\mathrm{d}x}\cos(\alpha x)\right] \\
&= \frac{\alpha(\cos(\alpha x))^2 + \alpha(\sin(\alpha x))^2}{(\cos(\alpha x))^2},
\end{aligned}
$$

where we have used $d\sin(\alpha x)/\mathrm{d}x = \alpha\cos(\alpha x)$ and $d\cos(\alpha x)/\mathrm{d}x = -\alpha\sin(\alpha x)$ from Example 3.10. Using the trigonometric identity $(\cos(\alpha x))^2 + (\sin(\alpha x))^2 = 1$ then shows that

$$
\frac{\mathrm{d}}{\mathrm{d}x}\tan(\alpha x) = \alpha\frac{1}{(\cos(\alpha x))^2} = \alpha(\sec(\alpha x))^2.
$$

It is left as an exercise to show that, on their domain of definition, we also have

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\cot(\alpha x) &= -\alpha(\csc(\alpha x))^2, \\
\frac{\mathrm{d}}{\mathrm{d}x}\sec(\alpha x) &= \alpha\sec(\alpha x)\tan(\alpha x), \\
\frac{\mathrm{d}}{\mathrm{d}x}\csc(\alpha x) &= -\alpha\csc(\alpha x)\cot(\alpha x).
\end{aligned}
$$

Example 3.16. Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = \sinh(\alpha x)/\cosh(\alpha x) = \tanh(\alpha x)$ for some $\alpha \in \mathbb{R} \setminus \{0\}$. By the quotient rule,

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\tanh(\alpha x) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\sinh(\alpha x)}{\cosh(\alpha x)}\right) \\
&= \frac{1}{(\cosh(\alpha x))^2}\left[\cosh(\alpha x)\frac{\mathrm{d}}{\mathrm{d}x}\sinh(\alpha x) - \sinh(\alpha x)\frac{\mathrm{d}}{\mathrm{d}x}\cosh(\alpha x)\right] \\
&= \frac{\alpha(\cosh(\alpha x))^2 - \alpha(\sinh(\alpha x))^2}{(\cosh(\alpha x))^2},
\end{aligned}
$$

where we have used $d\sinh(\alpha x)/\mathrm{d}x = \alpha\cosh(\alpha x)$ and $d\cosh(\alpha x)/\mathrm{d}x = \alpha\sinh(\alpha x)$ from Example 3.13. Using the hyperbolic identity $(\cosh(\alpha x))^2 - (\sinh(\alpha x))^2 = 1$ then shows that

$$\frac{\mathrm{d}}{\mathrm{d}x}\tanh(\alpha x) = \alpha\frac{1}{(\cosh(\alpha x))^2} = \alpha(\mathrm{sech}(\alpha x))^2.$$

It is left as an exercise to show that, on their domains of definition, we also have

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{coth}(\alpha x) &= -\alpha(\mathrm{csch}(\alpha x))^2,\\
\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{sech}(\alpha x) &= -\alpha\mathrm{sech}(\alpha x)\tanh(\alpha x),\\
\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{csch}(\alpha x) &= -\alpha\mathrm{csch}(\alpha x)\mathrm{coth}(\alpha x).
\end{aligned}$$

Out of the rules, the chain rule is perhaps considered to be the most powerful because it allows us to extend differentiation to any elementary function that is formed as a composition of functions. Recall that $g \circ f$ is defined by $(g \circ f)(x) = g(f(x))$ for all x for which this makes sense. The chain rule (assuming differentiability) then says that $(g\circ f)'(x) = g'(f(x))f'(x)$, which is read as "the derivative of g composed with f evaluated at x is given by the derivative of g evaluated at $f(x)$ multiplied by the derivative of f evaluated at x". If $y = g(u)$ and $u = f(x)$, so $y = g(f(x))$, another way to write the chain rule is

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u}\cdot\frac{\mathrm{d}u}{\mathrm{d}x}.$$

Let's see this in practice for some simple compositions.

Example 3.17. Differentiate $y(x) = \sin(x^2)$ for all $x \in \mathbb{R}$.

Solution: We can write this as $y(x) = \sin(u(x))$ with $u(x) = x^2$, and the chain rule tells us that

$$y'(x) = \cos(u(x))u'(x) = \cos(x^2)2x.$$

□

Example 3.18. Differentiate $y(x) = \sin(\cos(\tan x))$ for all $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

Solution: In this case we apply the chain rule multiple times. First, let $y(x) = \sin(u(x))$ with $u(x) = \cos(\tan(x))$. Then

$$y'(x) = \cos(u(x))u'(x) = \cos(\cos(\tan(x)))u'(x).$$

Next, to evaluate $u'(x)$ take $u(x) = \cos(v(x))$ with $v(x) = \tan(x)$. Then

$$y'(x) = \cos(\cos(\tan(x)))(-\sin(v(x)))v'(x) = -\cos(\cos(\tan(x)))\sin(\tan(x))(\sec(x))^2.$$

□

As the previous example indicates, you can continue to apply the chain rule indefinitely– hence the name "chain" rule!

We now use the chain rule to obtain some results for remaining basic functions; namely, exponentials of arbitrary base and power functions of arbitrary power.

Example 3.19. Let $a \in (0, \infty)$ and $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = a^x$ for all $x \in \mathbb{R}$. We can write

$$f(x) = a^x = (a)^x = (e^{\ln(a)})^x = e^{x \ln(a)} \tag{3.3}$$

and apply the chain rule to $f(x) = e^{u(x)}$ with $u(x) = x \ln(a)$. We then have

$$f'(x) = e^{u(x)} u'(x) = e^{x \ln(a)} \ln(a) = a^x \ln(a).$$

for all $x \in \mathbb{R}$, as required.

Example 3.20. Let $\alpha \in \mathbb{R} \setminus \{0\}$ and $f : (0, \infty) \to \mathbb{R}$ be given by $f(x) = x^\alpha$. We can write

$$f(x) = x^\alpha = (e^{\ln(x)})^\alpha = e^{\alpha \ln(x)} \tag{3.4}$$

and apply the chain rule to $f(x) = e^{u(x)}$ with $u(x) = \alpha \ln(x)$. We then have

$$f'(x) = e^{u(x)} u'(x) = e^{\alpha \ln(x)} \frac{\alpha}{x} = \alpha x^{\alpha - 1}.$$

Note that this is stated in Result 3.8 and may hold on a larger domain (depending on α), but the method used here requires $x > 0$ to evaluate the logarithm.

As a final rule, which we separate from those above, we provide the following result to allow differentiation of inverse functions and prove it using a straightforward application of the chain rule.

Result 3.21. (Derivative of an inverse function). Suppose that f is a bijective, differentiable function, so that its inverse function f^{-1} exists. Let $y = f(x)$, and suppose that f^{-1} is differentiable at y. If $f'(x) \neq 0$, then

$$(f^{-1})'(y) = \frac{1}{f'(x)}.$$

Equivalently,

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}.$$

Proof. Since $f^{-1}(f(x)) = x$, the chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[f^{-1}(f(x))\right] = (f^{-1})'(f(x))f'(x) = 1.$$

Since $f'(x) \neq 0$, we can divide by $f'(x)$ to obtain

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)}.$$

Writing $y = f(x)$ gives

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}.$$

□

Example 3.22. Let

$$f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to [-1, 1]$$

be given by $f(x) = \sin(x)$. The function f is bijective and differentiable at every point in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Its inverse function

$$f^{-1} : [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

is given by

$$f^{-1}(y) = \arcsin(y).$$

Using Result 3.21 (technically, we have to show that arcsin is differentiable at all $y \in (-1, 1)$, but we simply assume this here), we have

$$(f^{-1})'(y) = \frac{1}{f'(x)} = \frac{1}{\cos(x)},$$

which is well-defined for $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ since we then have $\cos(x) > 0$. Therefore, using the trigonometric identity

$$(\cos(x))^2 + (\sin(x))^2 = 1,$$

we obtain

$$(f^{-1})'(y) = \frac{1}{\sqrt{1 - (\sin(x))^2}}.$$

Since $x = \arcsin(y)$, we can substitute to obtain

$$(f^{-1})'(y) = \frac{1}{\sqrt{1 - (\sin(\arcsin(y)))^2}} = \frac{1}{\sqrt{1 - y^2}}.$$

Thus,

$$\frac{d}{dy}\arcsin(y) = \frac{1}{\sqrt{1 - y^2}} \quad \text{for all} \quad y \in (-1, 1).$$

We end this section with a summary of the formulae for the derivatives of basic functions (with the domain convention applying), and leave it as an exercise to prove those that have not already been covered above.

$$
\begin{aligned}
&\frac{\mathrm{d}}{\mathrm{d}x}e^x = e^x\,, && \frac{\mathrm{d}}{\mathrm{d}x}a^x = a^x\ln(a) \quad a>0\,,\\
&\frac{\mathrm{d}}{\mathrm{d}x}\ln(x) = \frac{1}{x}\,, && \frac{\mathrm{d}}{\mathrm{d}x}\log_a(x) = \frac{1}{x\ln(a)} \quad a>0, a\neq 1\,,\\
&\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \cos(x)\,, && \frac{\mathrm{d}}{\mathrm{d}x}\cos(x) = -\sin(x)\,,\\
&\frac{\mathrm{d}}{\mathrm{d}x}\tan(x) = (\sec(x))^2\,, && \frac{\mathrm{d}}{\mathrm{d}x}\cot(x) = -(\csc(x))^2\,,\\
&\frac{\mathrm{d}}{\mathrm{d}x}\sec(x) = \sec(x)\tan(x)\,, && \frac{\mathrm{d}}{\mathrm{d}x}\csc(x) = -\csc(x)\cot(x)\,,\\
&\frac{\mathrm{d}}{\mathrm{d}x}\arcsin(x) = \frac{1}{\sqrt{1-x^2}}\,, && \frac{\mathrm{d}}{\mathrm{d}x}\arccos(x) = -\frac{1}{\sqrt{1-x^2}}\,,\\
&\frac{\mathrm{d}}{\mathrm{d}x}\arctan(x) = \frac{1}{1+x^2}\,, && \frac{\mathrm{d}}{\mathrm{d}x}\mathrm{arccot}(x) = -\frac{1}{1+x^2}\,,\\
&\frac{\mathrm{d}}{\mathrm{d}x}\sinh(x) = \cosh(x)\,, && \frac{\mathrm{d}}{\mathrm{d}x}\cosh(x) = \sinh(x)\,,\\
&\frac{\mathrm{d}}{\mathrm{d}x}\tanh(x) = \mathrm{sech}^2(x)\,, && \frac{\mathrm{d}}{\mathrm{d}x}\coth(x) = -\,\mathrm{csch}^2(x)\,,\\
&\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{sech}(x) = -\,\mathrm{sech}(x)\tanh(x)\,, && \frac{\mathrm{d}}{\mathrm{d}x}\mathrm{csch}(x) = -\,\mathrm{csch}(x)\coth(x)\,,\\
&\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{arcsinh}(x) = \frac{1}{\sqrt{1+x^2}}\,, && \frac{\mathrm{d}}{\mathrm{d}x}\mathrm{arccosh}(x) = \frac{1}{\sqrt{x^2-1}}\,,\\
&\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{arctanh}(x) = \frac{1}{1-x^2}\,, && \frac{\mathrm{d}}{\mathrm{d}x}\mathrm{arccoth}(x) = \frac{1}{1-x^2}\,,\\
&\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{arcsech}(x) = -\frac{1}{x\sqrt{1-x^2}}\,, && \frac{\mathrm{d}}{\mathrm{d}x}\mathrm{arccsch}(x) = -\frac{1}{|x|\sqrt{1+x^2}}\,.
\end{aligned}
\tag{3.5}
$$

3.3 Implicit and logarithmic differentiation

In this final section we provide two methods that enable us to tackle functions (or equations) that, at first, may look quite daunting to differentiate. We shall see that simple application of the rules of differentiation combined with a few small tricks expand the differentiation toolkit to such instances too.

3.3.1 Implicit differentiation

The functions we have seen so far can be described by expressing one variable *explicitly* in terms of another variable, in the general form $y = f(x)$ (for example, $y = x^2$ or

$y = \sin(x^2)$). Some functions are instead defined *implicitly* by a relation between x and y, such as:

$$x^2 + y^2 = 25 \quad \text{or} \quad x^3 + y^3 = 6xy.$$

In some cases it is possible to solve such an equation for y as an explicit function (or several functions) of x. For instance, if we solve $x^2+y^2 = 25$ for y, we get $y = \pm\sqrt{25 - x^2}$, so two of the functions determined by the implicit equation are $f_1(x) = \sqrt{25 - x^2}$ and $f_2(x) = -\sqrt{25 - x^2}$. However, it is not easy to solve $x^3 + y^3 = 6xy$ for y explicitly as a function of x, by hand. If we cannot solve for y, can we expect to differentiate to obtain y'? In some sense we still can.

When we say that f is a function defined implicitly by $x^3 + y^3 = 6xy$, what we actually mean is that the equation

$$x^3 + (f(x))^3 = 6xf(x)$$

is true for all values of x in the domain of f. Fortunately, we do not need to solve an equation for f (or y in terms of x) in order to find the derivative f' (or y'). Instead we can use the method of implicit differentiation. This consists of differentiating both sides of an equation with respect to x and then solving the resulting equation for y' in terms of x and y. In the examples in this section it is always assumed that the given equation determines y implicitly as a differentiable function of x so that the method of implicit differentiation can be applied.

Example 3.23. Find y' if $x^3 + y^3 = 6xy$.

Solution: We differentiate both sides of $x^3 + y^3 = 6xy$ with respect to x, considering y to be a function of x (i.e. $y(x)$). Using the chain rule on y^3, and the product rule on $6xy$, we obtain

$$3x^2 + 3y^2y' = 6xy' + 6y$$

and divide by 3 to give

$$x^2 + y^2y' = 2xy' + 2y\,.$$

We now solve for y', to find that

$$\begin{aligned} y^2y' - 2xy' &= 2y - x^2 \\ \left(y^2 - 2x\right)y' &= 2y - x^2 \\ y' &= \frac{2y - x^2}{y^2 - 2x}. \end{aligned}$$

□

Example 3.24. Find y' if $\sin(x + y) = y^2\cos(x)$.

Solution: Differentiating both sides implicitly with respect to x while using the chain and product rules we find that

$$\cos(x+y)\cdot(1+y') = y^2(-\sin(x)) + \cos(x)\,(2yy')\,.$$

Then, solving for y' gives

$$y' = \frac{y^2\sin x + \cos(x+y)}{2y\cos x - \cos(x+y)}\,,$$

as required. □

3.3.2 Logarithmic differentiation

The calculation of derivatives of complicated functions involving products, quotients, or powers can often be simplified by taking logarithms. The method used in the following examples is called logarithmic differentiation.

Example 3.25. Differentiate the following equation with respect to $x > 0$:

$$y = \frac{x^{3/4}\sqrt{x^2+1}}{(3x+2)^5}\,.$$

Solution: Since $x > 0$, we have $y > 0$, and hence we may take logarithms of both sides. Using the properties of logarithms, we obtain

$$\ln(y) = \frac{3}{4}\ln(x) + \frac{1}{2}\ln(x^2+1) - 5\ln(3x+2).$$

Now, differentiating both sides implicitly with respect to x gives

$$\frac{1}{y}y' = \frac{3}{4}\cdot\frac{1}{x} + \frac{1}{2}\cdot\frac{2x}{x^2+1} - 5\cdot\frac{3}{3x+2}\,.$$

Finally, solving for y' and substituting the expression for y, gives

$$y' = y\left(\frac{3}{4x} + \frac{x}{x^2+1} - \frac{15}{3x+2}\right) = \frac{x^{3/4}\sqrt{x^2+1}}{(3x+2)^5}\left(\frac{3}{4x} + \frac{x}{x^2+1} - \frac{15}{3x+2}\right).$$

□

Example 3.26. Differentiate the following equation with respect to $x > 0$:

$$y = x^{\sqrt{x}}\,. \tag{3.6}$$

Solution: Since both the base and the exponent are variables, we use logarithmic differentiation. Taking logarithms, gives

$$\ln(y) = \ln(x^{\sqrt{x}}) = \sqrt{x}\ln(x). \tag{3.7}$$

Implicit differentiation then gives

$$\frac{y'}{y} = \frac{\sqrt{x}}{x} + \frac{\ln(x)}{2\sqrt{x}},$$

which after rearranging, simplifying and substituting for y gives

$$y' = y\left(\frac{1}{\sqrt{x}} + \frac{\ln(x)}{2\sqrt{x}}\right) = x^{\sqrt{x}}\left(\frac{2+\ln(x)}{2\sqrt{x}}\right).$$

Note that another way to answer this question would be to write $y = e^{\sqrt{x}\ln(x)}$ and apply the chain rule. □

Chapter 4

Integration

4.1 Integrals and antiderivatives

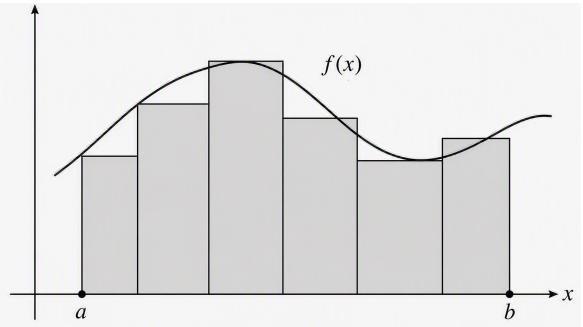
If you already saw differentiation prior to taking this course, it is likely you came across the concept of integration as well. A precise definition of integration (there are several) requires some more advanced real analysis that you will probably not see until the second year. Thus, we begin this section with a conceptual picture of the framework before turning to some concrete definitions and results that we can immediately apply in calculations.

4.1.1 Area under the curve

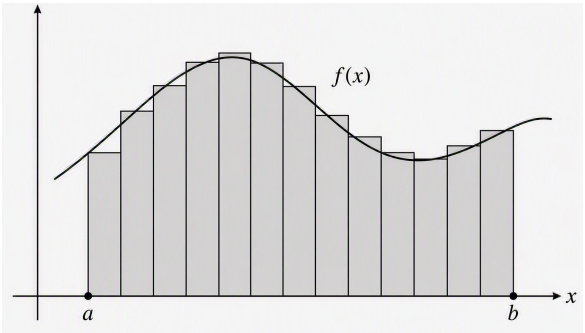
While the derivative is the slope of the tangent line to a curve (graph) of a function at a given point, integration is motivated by trying to calculate the *area under the curve (graph) over a given interval*. The basic idea is simple and dates back as far as the 4th Century BC[1]. Given a non-negative real function, you can try partitioning the area into individual regions for which the area or volume is known or easy to calculate (e.g. rectangular regions as in Figure 4.1a and 4.1b). By partitioning the interval into increasingly small regions and taking a suitable limit as the widths of those regions tend to zero, we can hope to obtain an exact value.

Archimedes used this concept to calculate the areas and volumes of various shapes in the 3rd Century BC, and similar methods were developed independently in China in the 3rd Century AD. This was extended to more complicated shapes by the mathematician Ibn al-Haytham. Major advances then appeared in the 17th Century AD when the theory of calculus was born, but the theory of integration did not really become formalised rigorously using limits until Riemann (circa 1854). However, Riemann's theory of integration

[1] Method of exhaustion of the ancient Greek mathematicians Democritus and Eudoxus (4th Century BC).



(a) Partitioning area under curve $y = f(x)$ into rectangular regions. Fewer than in panel (b).



(b) Partitioning area under curve $y = f(x)$ into rectangular regions. More than in panel (a).

did not extend to all functions and a different definition was formulated by Lebesgue to deal with more general cases (for example, the Dirichlet function is not Riemann integrable but it is Lebesgue integrable). Lebesgue integration is a more general theory of integration that plays a central role in modern analysis, but its definition requires measure theory and is beyond the scope of this course.

In general, the (definite) integral of a real-valued function $f(x)$ with respect to a real variable x on a closed interval $[a, b]$ where $a, b \in \mathbb{R}$ with $b > a$ is written as

$$\int_a^b f(x)\mathrm{d}x.$$

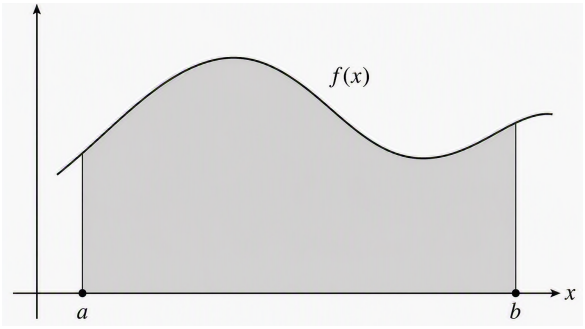
The symbol $\mathrm{d}x$, called the differential of the variable x, indicates that the variable of integration is x. The function $f(x)$ is called the integrand, the points a and b are called the limits (or bounds) of integration, and the integral is said to be over the interval $[a, b]$, called the interval of integration. A function is said to be integrable if its integral is well-defined and finite, which depends on the particular definition you are using (e.g. either Riemann or Lebesgue). As indicated, this gives the area between the curve (graph) of a non-negative function $f(x)$ and the x-axis over the interval $[a, b]$ (Figure 4.2a). If the curve dips below the x-axis, the integral gives a signed area. This means the integral adds the part above the x-axis as positive and subtracts the part below the x-axis as negative (Figure 4.2b). So, the result can be positive, negative, or zero, depending on how much of the curve is above or below the x-axis.

This geometric interpretation of the integral as a signed area gives us the following result: for every $a, b, c \in \mathbb{R}$ such that $a < c < b$ we have

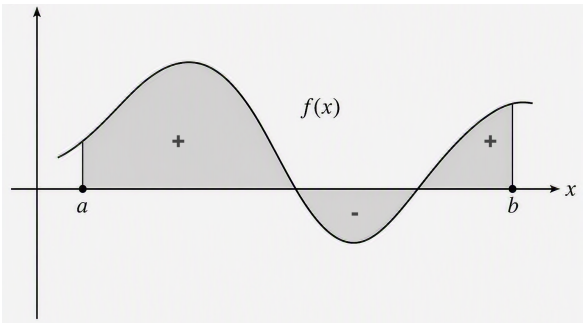
$$\int_a^b f(x)\mathrm{d}x = \int_c^b f(x)\mathrm{d}x + \int_a^c f(x)\mathrm{d}x.$$

Moreover, for any two continuous functions $f, g : [a, b] \to \mathbb{R}$ we have

$$\int_a^b [f(x) + g(x)]\mathrm{d}x = \int_a^b f(x)\mathrm{d}x + \int_a^b g(x)\mathrm{d}x.$$



(a) Shaded region representing area under curve of function $f(x)$ over interval $[a, b]$.



(b) Positive (labelled "+") and negative (labelled "−") contributions to area.

We also use the convention

$$\int_b^a f(x)\mathrm{d}x = -\int_a^b f(x)\mathrm{d}x.$$

Notice that we can replace the variable of integration with any other notation (x is sometimes called a "dummy variable") so that

$$\int_a^b f(x)\mathrm{d}x = \int_a^b f(u)\mathrm{d}u = ...$$

This means we can let a be fixed and allow the right endpoint to vary between a and b to obtain a function $F : [a, b] \to \mathbb{R}$ given by

$$F(x) = \int_a^x f(u)\mathrm{d}u.$$

That is to say, we have used x for the right endpoint of the interval and F is therefore the function that captures how the signed area under the curve on the interval $[a, x]$ varies with $x \in [a, b]$. You can think about F as capturing an "accumulating area" associated with f and the interval $[a, b]$.

4.1.2 The Fundamental Theorem of Calculus

It turns out that integration and differentiation are operations that effectively undo each other, which will be captured by the so-called Fundamental Theorem of Calculus that is split into two parts: the first part allows us to differentiate certain integral expressions whilst the second part allows us to compute integrals using so-called *antiderivatives*.

Definition 4.1. Given a function $f : (a, b) \to \mathbb{R}$ we say that a function $F : (a, b) \to \mathbb{R}$ is an *antiderivative* of f if F is differentiable and

$$F'(x) = f(x) \quad \text{for all } x \in (a, b).$$

Example 4.2. Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = x^n$ for some $n \in \mathbb{N}$. Then any function $F : \mathbb{R} \to \mathbb{R}$ given by

$$F(x) = \frac{x^{n+1}}{n+1} + C$$

where $C \in \mathbb{R}$ is any real constant provides an antiderivative of f. To see this note that $F'(x) = x^n + 0 = f(x)$ for all $x \in \mathbb{R}$. In fact, $x^{\alpha+1}/(\alpha+1)$ is an antiderivative of x^α for any $\alpha \in \mathbb{R} \setminus \{-1\}$, on the appropriate domain (see Result 3.8 from the previous section).

The above example shows that antiderivatives are not defined uniquely, but for any antiderivative F_1 of f we can always find another, F_2, by adding on a constant. Indeed, if F_1 and F_2 are two antiderivatives of the same function $f : (a, b) \to \mathbb{R}$, then there exists a constant $C \in \mathbb{R}$ such that $F_1(x) - F_2(x) = C$ for all $x \in (a, b)$. The name antiderivative captures the concept that integration and differentiation are effective inverse operations, as encompassed by the first part of the Fundamental Theorem of Calculus.

Result 4.3. (First Fundamental Theorem of Calculus). Let $f : [a, b] \to \mathbb{R}$ be a continuous real function and $F : [a, b] \to \mathbb{R}$ be defined for all $x \in [a, b]$ by

$$F(x) = \int_a^x f(u)du.$$

Then F is continuous on $[a, b]$ and differentiable on (a, b) with $F'(x) = f(x)$ for all $x \in (a, b)$.

It is worth remarking that the assumptions on f (continuity on $[a, b]$) imply that it is integrable[2] and so this ensures F is well-defined. The following result is the continuous-function version of the Second Fundamental Theorem of Calculus[3].

Result 4.4. (Second Fundamental Theorem of Calculus, continuous case). Let $f : [a, b] \to \mathbb{R}$ be a continuous real function and $F : [a, b] \to \mathbb{R}$ be a continuous function that is an antiderivative of f on (a, b). That is, $F'(x) = f(x)$ for all $x \in (a, b)$. Then

$$\int_a^b f(x)\mathrm{d}x = F(b) - F(a).$$

The Fundamental Theorem of Calculus (both parts combined) says that integration and differentiation undo each other.

[2]Technically, this is enough to ensure f is Riemann and therefore also Lebesgue integrable.

[3]The general form of the Second Fundamental Theorem of Calculus only assumes f is integrable on $[a, b]$.

We introduce alternative notation for the family of antiderivatives of f, namely using the integral symbol with no bounds, which is also called the *indefinite integral* of f:

$$\int f(x)\mathrm{d}x = F(x) + C, \quad C \in \mathbb{R} \tag{4.1}$$

(so F is *an* antiderivative rather than *the* indefinite integral). However, the Second Fundamental Theorem of Calculus shows that to evaluate definite integrals only one antiderivative of f is required since cancellation of the undetermined constants will occur, i.e. $(F(b) + C) - (F(a) + C) = F(b) - F(a)$. Thus, given any antiderivative $F(x)$ we can evaluate the integral as follows:

$$\int_a^b f(x)\mathrm{d}x = F(b) - F(a) \equiv [F(x)]_a^b.$$

Example 4.5. The function $F : [-\pi, \pi] \to \mathbb{R}$ given by $F(x) = \sin(x)$ is an antiderivative of the function $f(x) = \cos(x)$ since $F'(x) = \cos(x) = f(x)$. We then have

$$\int_{-\pi}^{\pi} \cos(x)\mathrm{d}x = [F(x)]_{-\pi}^{\pi} = [\sin(x)]_{-\pi}^{\pi} = \sin(\pi) - \sin(-\pi) = 0 - 0 = 0.$$

Example 4.6. The function $F : [0, 1] \to \mathbb{R}$ given by $F(x) = x^2$ is an antiderivative of the function $f(x) = 2x$ since $F'(x) = 2x = f(x)$. We then have

$$\int_0^1 2x\mathrm{d}x = [x^2]_0^1 = 1^2 - 0 = 1.$$

4.2 Integral calculus

The Fundamental Theorem of Calculus provides a bridge connecting differentiation and integration. We will use it to establish the calculus techniques for integrating products and compositions of functions using Integration by Parts and the Substitution Formula. We will also see how the indefinite integral of any rational function can, in principle, be found using the method of Partial Fractions.

4.2.1 Integration by Parts

We can combine the Fundamental Theorem of Calculus with the product rule for derivatives to obtain the following result for definite and indefinite integrals of products of functions.

Result 4.7. (Integration by Parts) If $u, v : [a, b] \to \mathbb{R}$ are differentiable with continuous derivatives $u', v' : [a, b] \to \mathbb{R}$, then

$$\int_a^b u(x)v'(x)\mathrm{d}x = \big[u(x)v(x)\big]_a^b - \int_a^b u'(x)v(x)\mathrm{d}x$$

and

$$\int u(x)v'(x)\mathrm{d}x = u(x)v(x) - \int u'(x)v(x)\mathrm{d}x$$

where we recall $\big[u(x)v(x)\big]_a^b = u(b)v(b) - u(a)v(a)$.

Note that the product rule gives $(uv)' = u'v + uv'$ and rearranging to get $uv' = (uv)' - u'v$, followed by integrating both sides and applying the Fundamental Theorem of Calculus to $(uv)'$, gives a sketch proof of this result. The second statement on indefinite integrals in this result can be extended to include differentiable functions with continuous derivatives defined on open intervals as well.

The next example shows some useful strategies for obtaining integrals and indefinite integrals using Integration by Parts. In these examples, we are using previous results from the sections on differentiation to identify the appropriate antiderivative in each case.

Example 4.8. Find the integral $\displaystyle\int_0^{\frac{\pi}{2}} x\cos(x)\ \mathrm{d}x$.

Solution: We apply Integration by Parts with

$$\begin{cases} u(x) = x \\ v(x) = \sin(x) \end{cases} \quad \text{and} \quad \begin{cases} u'(x) = 1 \\ v'(x) = \cos(x) \end{cases}$$

to obtain

$$\int_0^{\frac{\pi}{2}} x\cos(x)\ \mathrm{d}x = [x\sin(x)]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin(x)\ \mathrm{d}x = \tfrac{\pi}{2} - [-\cos(x)]_0^{\frac{\pi}{2}} = \tfrac{\pi}{2} - 1.$$

In this case we have used the fact that $-\cos(x)$ is an antiderivative of $\sin(x)$. □

Example 4.9. Find the indefinite integral of $f : (0, \infty) \to \mathbb{R}$, given by $f(x) = \ln(x)$ for all $x \in (0, \infty)$.

Solution: We apply Integration by Parts with

$$\begin{cases} u(x) = \ln(x) \\ v(x) = x \end{cases} \quad \text{and} \quad \begin{cases} u'(x) = \frac{1}{x} \\ v'(x) = 1 \end{cases}$$

to obtain

$$\int \ln(x)\ \mathrm{d}x = x\ln(x) - \int \left(\frac{1}{x}\right) x\ \mathrm{d}x = x\ln(x) - x + C, \quad \text{for } C \in \mathbb{R}.$$

In this case we have used the fact that x is an antiderivative of 1. □

Example 4.10. Find the indefinite integral $\int x^2 e^x \, \mathrm{d}x$.

Solution: We apply Integration by Parts with

$$\begin{cases} u(x) = x^2 \\ v(x) = e^x \end{cases} \quad \text{and} \quad \begin{cases} u'(x) = 2x \\ v'(x) = e^x \end{cases}$$

to obtain

$$\int x^2 e^x \, \mathrm{d}x = x^2 e^x - \int 2x e^x \, \mathrm{d}x.$$

Applying integration by Parts again to the indefinite integral on the right hand side with

$$\begin{cases} u(x) = x \\ v(x) = e^x \end{cases} \quad \text{and} \quad \begin{cases} u'(x) = 1 \\ v'(x) = e^x \end{cases}$$

gives

$$\int x e^x \, \mathrm{d}x = x e^x - \int e^x \, \mathrm{d}x = x e^x - e^x + C = (x-1)e^x + C, \quad \text{for } C \in \mathbb{R}.$$

In this case we have used the fact that e^x is an antiderivative of itself. Altogether, we therefore have

$$\int x^2 e^x \, \mathrm{d}x = x^2 e^x - 2(x-1)e^x = (x^2 - 2x + 2)e^x + C, \quad \text{for } C \in \mathbb{R}.$$

□

Observe that in the previous example we were free to bring the constant factor of 2 through the integral sign. More generally, if $c \in \mathbb{R}$ is a constant and f is a function with an antiderivative, then

$$\int c f(x) \, \mathrm{d}x = c \int f(x) \, \mathrm{d}x.$$

Indeed, if F is an antiderivative of f, then cF is an antiderivative of cf, since

$$(cF)'(x) = cF'(x) = cf(x).$$

Thus, any constant factor may be taken outside an indefinite integral. Similarly, for definite integrals we have

$$\int_a^b c f(x) \, \mathrm{d}x = c \int_a^b f(x) \, \mathrm{d}x.$$

The previous example also showed that we may continue to apply Integration by Parts as many times as necessary.

Example 4.11. Find the indefinite integral $\int e^x \cos(x) \, \mathrm{d}x$.

Solution: We apply Integration by Parts with

$$\begin{cases} u(x) = e^x \\ v(x) = \sin(x) \end{cases} \quad \text{and} \quad \begin{cases} u'(x) = e^x \\ v'(x) = \cos(x) \end{cases}$$

to obtain

$$\int e^x \cos(x)\ \mathrm{d}x = e^x \sin(x) - \int e^x \sin(x)\ \mathrm{d}x.$$

Applying Integration by Parts again with

$$\begin{cases} u(x) = e^x \\ v(x) = -\cos(x) \end{cases} \quad \text{and} \quad \begin{cases} u'(x) = e^x \\ v'(x) = \sin(x) \end{cases}$$

gives

$$\int e^x \sin(x)\ \mathrm{d}x = -e^x \cos(x) + \int e^x \cos(x)\ \mathrm{d}x.$$

Altogether, we have

$$\int e^x \cos(x)\ \mathrm{d}x = e^x \sin(x) - \left(-e^x \cos(x) + \int e^x \cos(x)\ \mathrm{d}x\right)$$

and after rearranging and including the constant of integration this becomes

$$\int e^x \cos(x)\ \mathrm{d}x = \frac{1}{2}(e^x \sin(x) + e^x \cos(x)) + C, \quad \text{for } C \in \mathbb{R}.$$

□

4.2.2 Substitution Formula

The following result provides a systematic method for evaluating certain integrals by substitution. The basic idea is to introduce a new variable $u = u(x)$ so that the integral is transformed into a simpler one.

Result 4.12. (Substitution Formula) If $f : \mathbb{R} \to \mathbb{R}$ is a continuous function[4] and $u : [a, b] \to \mathbb{R}$ is differentiable with a continuous derivative $u' : [a, b] \to \mathbb{R}$, then

$$\int_a^b f(u(x))u'(x)\ \mathrm{d}x = \int_{u(a)}^{u(b)} f(y)\ \mathrm{d}y.$$

Notice that in Result 4.12 we use y as a dummy variable on the right hand side so as to avoid confusion with x and u. The following example shows how to apply this result.

[4] In fact f only needs to be defined and continuous on the range of u.

Example 4.13. Find the value of $\int_1^2 e^{-3x^4+1}x^3 \, \mathrm{d}x$.

Solution: Let $f(x) = e^x$ and $u(x) = -3x^4 + 1$ for all $x \in [1, 2]$, so $u'(x) = -12x^3$, $u(1) = -2$ and $u(2) = -47$. We see that the assumptions of Result 4.12 hold, and thus we obtain

$$\begin{aligned}\int_1^2 e^{-3x^4+1}x^3 \, \mathrm{d}x &= -\frac{1}{12}\int_1^2 e^{u(x)}u'(x) \, \mathrm{d}x \\ &= -\frac{1}{12}\int_{-2}^{-47} e^y \, \mathrm{d}y \\ &= \frac{1}{12}\int_{-47}^{-2} e^y \, \mathrm{d}y \\ &= \frac{1}{12}(e^{-2} - e^{-47}).\end{aligned}$$

Again we use the fact that e^y is an antiderivative of itself. □

Although the proof of Result 4.12 is not provided here[5], a useful consequence is that we can interpret the notation used as saying that

$$u'(x)\mathrm{d}x = \frac{\mathrm{d}u}{\mathrm{d}x}\mathrm{d}x = \mathrm{d}u.$$

Thus, for indefinite integrals we could write this result as

$$\int f(u(x))u'(x) \, \mathrm{d}x = \int f(u) \, \mathrm{d}u.$$

Example 4.14. Compute $\int x\cos(x^2 + 1)\mathrm{d}x$.

Solution: Let $u = x^2 + 1$ so that $u'(x) = 2x$ and thus $\mathrm{d}u = 2x\mathrm{d}x$. Then

$$\int x\cos(x^2 + 1)\mathrm{d}x = \frac{1}{2}\int \cos(u)\mathrm{d}u = \frac{1}{2}\sin(u) + C = \frac{1}{2}\sin(x^2 + 1) + C \quad \text{for } C \in \mathbb{R},$$

where we have used that $\sin(x)$ is an antiderivative of $\cos(x)$. □

When evaluating definite integrals by substitution, there are two standard approaches: (1) transform the limits of integration into u-values and evaluate directly, or (2) find the general antiderivative in terms of x via indefinite integration and apply the original x-limits.

[5]It may be put on a rigorous foundation by interpreting it as a statement about differential forms.

Example 4.15. Consider the integral

$$\int_0^2 \frac{x}{x^2+1}\mathrm{d}x.$$

We again make the substitution $u = x^2+1$ so that $\mathrm{d}u = 2x\mathrm{d}x$. In the first case, this gives

$$\int_0^2 \frac{x}{x^2+1}\mathrm{d}x = \frac{1}{2}\int_{0^2+1}^{2^2+1} \frac{1}{u}\mathrm{d}u = \frac{1}{2}\int_1^5 \frac{1}{u}\mathrm{d}u = \frac{1}{2}[\ln(5)-\ln(1)] = \frac{\ln(5)}{2},$$

where we have used the fact that $\ln(u)$ is an antiderivative of $1/u$ when $u > 0$ (recall $u = x^2+1$ with $x \in [0,2]$). In the second case, we first find the indefinite integral via substitution

$$\int \frac{x}{x^2+1}\mathrm{d}x = \frac{1}{2}\int \frac{1}{u}\mathrm{d}u = \frac{\ln(u)}{2} + C = \frac{\ln(x^2+1)}{2} + C \quad \text{for } C \in \mathbb{R},$$

and then evaluate at bounds to give

$$\int_0^2 \frac{x}{x^2+1}\mathrm{d}x = \frac{\ln(2^2+1)}{2} - \frac{\ln(0^2+1)}{2} = \frac{\ln(5)}{2}.$$

Either approach is absolutely fine, provided the indefinite integral or antiderivative makes sense over the domain of interest.

In many cases, substitution combined with the application of an appropriate trigonometric identity can prove a powerful tool when integrating combinations of trigonometric functions.

Example 4.16. Compute $\int (\cos(x))^3 \, \mathrm{d}x$.

Solution: We substitute $u(x) = \sin(x)$, noting that $u'(x) = \cos(x)$ so that $\mathrm{d}u = \cos(x)\mathrm{d}x$, to obtain

$$\begin{aligned}
\int (\cos(x))^3 \, \mathrm{d}x &= \int (\cos(x))^2 \cos(x) \, \mathrm{d}x \\
&= \int (1-(\sin(x))^2)\cos(x) \, \mathrm{d}x \\
&= \int (1-u^2) \, \mathrm{d}u \\
&= u - \frac{1}{3}u^3 + C \\
&= \sin(x) - \frac{1}{3}(\sin(x))^3 + C, \quad \text{for } C \in \mathbb{R},
\end{aligned}$$

where in going from first to second line we used the identity $(\cos(x))^2 + (\sin(x))^2 = 1$ and in the second from last line used $u - u^3/3$ as an antiderivative of $1-u^2$. □

Example 4.17. Compute $\displaystyle\int \frac{(\sin(x))^7}{(\cos(x))^4}\,\mathrm{d}x$.

Solution: We substitute $u(x) = \cos(x)$, noting that $u'(x) = -\sin(x)$ so that $\mathrm{d}u = -\sin(x)\mathrm{d}x$, to obtain

$$\begin{aligned}
\int \frac{(\sin(x))^7}{(\cos(x))^4}\,\mathrm{d}x &= \int (\sin(x))^6(\cos(x))^{-4}\sin(x)\,\mathrm{d}x \\
&= -\int (1-(\cos(x))^2)^3(\cos(x))^{-4}(-\sin(x))\,\mathrm{d}x \\
&= -\int (1-u^2)^3u^{-4}\,\mathrm{d}u \\
&= -\int (1-3u^2+3u^4-u^6)u^{-4}\,\mathrm{d}u \\
&= \int (-u^{-4}+3u^{-2}-3+u^2)\,\mathrm{d}u \\
&= \frac{1}{3u^3}-\frac{3}{u}-3u+\frac{1}{3}u^3+C \\
&= \frac{1}{3\cos^3(x)}-\frac{3}{\cos(x)}-3\cos(x)+\tfrac{1}{3}\cos^3(x)+C, \quad \text{for } C\in\mathbb{R}.
\end{aligned}$$

Here we have used the fact that $x^{\alpha+1}/(\alpha+1)$ is an antiderivative of x^α for any $\alpha \in \mathbb{R}\backslash\{-1\}$, on the appropriate domain. □

Example 4.18. Compute $\displaystyle\int \cos^4(x)\,\mathrm{d}x$.

Solution: This example does not actually require a substitution, but we include it here for completeness (see the next example). Instead, we use a double-angle identity $(\cos(x))^2 = (1+\cos(2x))/2$ twice to obtain

$$\begin{aligned}
\int \cos^4(x)\,\mathrm{d}x &= \int \left(\frac{1}{2}(1+\cos(2x))\right)^2\,\mathrm{d}x \\
&= \frac{1}{4}\int \left(1+2\cos(2x)+(\cos(2x))^2\right)\,\mathrm{d}x \\
&= \frac{1}{4}\int \left(1+2\cos(2x)+\frac{1}{2}(1+\cos(4x))\right)\,\mathrm{d}x \\
&= \frac{1}{4}\int \left(\frac{3}{2}+2\cos(2x)+\frac{1}{2}\cos(4x)\right)\,\mathrm{d}x \\
&= \frac{1}{4}\left(\frac{3}{2}x+2\frac{1}{2}\sin(2x)+\frac{1}{2}\frac{1}{4}\sin(4x)\right)+C \\
&= \frac{3}{8}x+\frac{1}{4}\sin(2x)+\frac{1}{32}\sin(4x)+C, \quad \text{for } C\in\mathbb{R}.
\end{aligned}$$

Here we have used the fact that $\sin(\alpha x)/\alpha$ is an antiderivative of $\cos(\alpha x)$. □

The examples above illustrate two useful approaches for integrating powers of trigonometric functions. When one of the powers of $\sin(x)$ or $\cos(x)$ is odd, we can often isolate a single factor and use substitution to reduce the integral to a polynomial antiderivative. When the relevant powers are even, we instead use trigonometric identities to reduce the powers.

Example 4.19. Compute $\int (\sin(x))^m (\cos(x))^n \, \mathrm{d}x$ for any natural numbers $n, m \in \mathbb{N}$.

Solution: If n is an odd number, then $n = 2k+1$ for some $k \in \mathbb{N} \cup \{0\}$, so we substitute $u(x) = \sin(x)$, noting that $u'(x) = \cos(x)$ so that $\mathrm{d}u = \cos(x)\mathrm{d}x$, to obtain

$$\begin{aligned}\int (\sin(x))^m (\cos(x))^{2k+1} \, \mathrm{d}x &= \int (\sin(x))^m (\cos(x))^{2k} \cos(x) \, \mathrm{d}x \\ &= \int (\sin(x))^m (1 - (\sin(x))^2)^k \cos(x) \, \mathrm{d}x \\ &= \int u^m (1-u^2)^k \, \mathrm{d}u,\end{aligned}$$

where we used $(\cos(x))^2 + (\sin(x))^2 = 1$ in the second line. This reduces matters to finding an antiderivative of a polynomial, which we can do by finding an antiderivative of each monomial as per Example 4.2.

If m is an odd number, then $m = 2k+1$ for some $k \in \mathbb{N} \cup \{0\}$, so we substitute $u(x) = \cos(x)$, noting that $u'(x) = -\sin(x)$, to obtain

$$\begin{aligned}\int (\sin(x))^{2k+1} (\cos(x))^n \, \mathrm{d}x &= \int (\sin(x))^{2k} (\cos(x))^n \sin(x) \, \mathrm{d}x \\ &= -\int (1 - (\cos(x))^2)^k (\cos(x))^n (-\sin(x)) \, \mathrm{d}x \\ &= -\int (1-u^2)^k u^n \, \mathrm{d}u,\end{aligned}$$

where again we used $(\cos(x))^2 + (\sin(x))^2 = 1$ in the second line. This again reduces matters to finding an antiderivative of a polynomial.

However, if n and m are both even numbers, there is no single factor of $\sin(x)$ or $\cos(x)$ naturally left over for such a substitution. In this case, the double-angle identities below can be used repeatedly to reduce the powers of $(\sin(x))^2$ or $(\cos(x))^2$:

$$(\sin(x))^2 = \tfrac{1}{2}(1 - \cos(2x)) \quad (\cos(x))^2 = \tfrac{1}{2}(1 + \cos(2x)). \tag{4.2}$$

□

We can use an approach that somewhat mirrors the relation between sine and cosine functions to find indefinite integrals of products of secant and tangent functions.

Example 4.20. Consider the family of indefinite integrals

$$\int (\tan(x))^m (\sec(x))^n \, \mathrm{d}x$$

for any natural numbers $n, m \in \mathbb{N}$.

If n is an even number, then $n = 2k$ for some $k \in \mathbb{N}$, so we substitute $u(x) = \tan(x)$, noting that $u'(x) = (\sec(x))^2$ so that $\mathrm{d}u = (\sec(x))^2 \mathrm{d}x$, to obtain

$$\begin{aligned}\int (\tan(x))^m (\sec(x))^{2k} \, \mathrm{d}x &= \int (\tan(x))^m (\sec(x))^{2k-2} (\sec(x))^2 \, \mathrm{d}x \\ &= \int (\tan(x))^m ((\tan(x))^2 + 1)^{k-1} (\sec(x))^2 \, \mathrm{d}x \\ &= \int u^m (u^2 + 1)^{k-1} \, \mathrm{d}u,\end{aligned}$$

where we used $(\sec(x))^2 - (\tan(x))^2 = 1$ in the second line. This reduces matters to finding an antiderivative of a polynomial.

If m is an odd number, then $m = 2k + 1$ for some $k \in \mathbb{N} \cup \{0\}$, so we substitute $u(x) = \sec(x)$, noting that $u'(x) = \sec(x)\tan(x)$ so that $\mathrm{d}u = \sec(x)\tan(x)\mathrm{d}x$, to obtain

$$\begin{aligned}\int (\tan(x))^{2k+1} (\sec(x))^n \, \mathrm{d}x &= \int (\tan(x))^{2k} (\sec(x))^{n-1} \sec(x)\tan(x) \, \mathrm{d}x \\ &= \int ((\sec(x))^2 - 1)^k (\sec(x))^{n-1} \sec(x)\tan(x) \, \mathrm{d}x \\ &= \int (u^2 - 1)^k u^{n-1} \, \mathrm{d}u,\end{aligned}$$

where we again used $(\sec(x))^2 - (\tan(x))^2 = 1$ in the second line. This again reduces matters to finding an antiderivative of a polynomial.

For the remaining cases from the previous example, where n is an odd number and m is an even number, there does not seem to be a general strategy that always works. We often need to think creatively, as the following examples show.

Example 4.21. Compute $\int \tan(x) \, \mathrm{d}x$.

Solution: We substitute $u(x) = \cos(x)$, noting that $u'(x) = -\sin(x)$ so that $\mathrm{d}u = -\sin(x)\mathrm{d}x$, to obtain

$$\int \tan(x) \, \mathrm{d}x = \int \frac{\sin(x)}{\cos(x)} \, \mathrm{d}x$$

$$\begin{aligned}
&= -\int \frac{1}{u}\,\mathrm{d}u \\
&= -\ln(|u|) + C \\
&= -\ln(|\cos(x)|) + C, \quad \text{for } C \in \mathbb{R}.
\end{aligned}$$

Here we have used the fact that $\ln(|u|)$ is an antiderivative of $1/u$ on the appropriate domain. □

Example 4.22. Compute $\int \sec(x)\,\mathrm{d}x$.

Solution: We substitute $u(x) = \sec(x) + \tan(x)$, noting that $u'(x) = \sec(x)\tan(x) + (\sec(x))^2$ so that $\mathrm{d}u = (\sec(x)\tan(x) + (\sec(x))^2)\mathrm{d}x$, to obtain

$$\begin{aligned}
\int \sec(x)\,\mathrm{d}x &= \int \sec(x)\frac{\sec(x)+\tan(x)}{\sec(x)+\tan(x)}\,\mathrm{d}x \\
&= \int \frac{(\sec(x))^2 + \sec(x)\tan(x)}{\sec(x)+\tan(x)}\,\mathrm{d}x \\
&= \int \frac{1}{u}\,\mathrm{d}u \\
&= \ln(|u|) + C \\
&= \ln(|\sec(x)+\tan(x)|) + C, \quad \text{for } C \in \mathbb{R}.
\end{aligned}$$

□

4.2.3 Method of Partial Fractions

Previously, using substitution we could compute indefinite integrals such as

$$\int \frac{2x-1}{x^2-x-6}\mathrm{d}x$$

since the substitution $u(x) = x^2 - x - 6$ gives

$$\int \frac{2x-1}{x^2-x-6}\mathrm{d}x = \int \frac{1}{u}\mathrm{d}u = \ln(|u|) + C = \ln(|x^2-x-6|) + C, \quad C \in \mathbb{R}.$$

In this case, the integrand is a *rational function*: that is, it is a function that can be written as the quotient of two polynomials. Also, the numerator is the derivative of the denominator, making the integral straightforward by substitution. However, the numerator of a rational function will not always be a constant multiple of the derivative of the denominator, so we need a more systematic technique.

Example 4.23. Consider the indefinite integral

$$\int \frac{x-3}{x^2-1}\mathrm{d}x.$$

In this case the numerator is not a constant multiple of the derivative of the denominator. However, addition of two fractions extends to rational functions such that, for any $A, B \in \mathbb{R}$, we have

$$\frac{A}{x-1}+\frac{B}{x+1}=\frac{A(x+1)+B(x-1)}{(x-1)(x+1)}.$$

Thus, factorising the denominator and working backwards, we notice that the integrand can be broken up as follows

$$\frac{x-3}{x^2-1}=\frac{x-3}{(x-1)(x+1)}=\frac{A}{x-1}+\frac{B}{x+1}$$

with some $A, B \in \mathbb{R}$ obtained by solving

$$A(x+1)+B(x-1)=(A+B)x+(A-B)=x-3.$$

Equivalently,

$$A+B=1, \quad A-B=-3,$$

which gives $A=-1$ and $B=2$. We can therefore write

$$\int \frac{x-3}{x^2-1}\mathrm{d}x=\int\left[-\frac{1}{x-1}+\frac{2}{x+1}\right]\mathrm{d}x=-\int\frac{1}{x-1}\mathrm{d}x+2\int\frac{1}{x+1}\mathrm{d}x,$$

and now use substitution on each term individually to obtain

$$\int \frac{x-3}{x^2-1}\mathrm{d}x=-\ln(|x-1|)+2\ln(|x+1|)+C=\ln\left(\frac{|x+1|^2}{|x-1|}\right)+C, \quad C\in\mathbb{R}.$$

This process of taking a rational function and decomposing it into simpler rational functions that we can add or subtract to get the original is called *partial fraction decomposition*. Many integrals involving rational functions can be done if we first do partial fraction decomposition on the integrand. The denominator determines the form of the partial fractions. For example, when the denominator contains two distinct linear factors,

$$(x-a)(x-b) \quad a,b\in\mathbb{R},$$

we try

$$\frac{A}{x-a}+\frac{B}{x-b}.$$

If a factor is repeated, then we need a term for each of its powers. For example,

$$(x-a)^2(x-b)$$

leads to

$$\frac{A}{x-a}+\frac{B}{(x-a)^2}+\frac{C}{x-b}.$$

The general result below summarises these patterns. It is important to note that, in the form described, this procedure can only be applied if the degree (largest power in the polynomial) of the numerator is strictly less than the degree of the denominator. However, this is not a serious restriction: we can often first rewrite a rational function as the sum of a polynomial and a rational function whose numerator has strictly smaller degree than its denominator.

Result 4.24. (Partial Fraction Decomposition). Let p and q be polynomials such that the degree of p is strictly smaller than the degree of q. Moreover, suppose that q is factorised over $\mathbb{R}$ into linear factors and irreducible quadratic factors. Then the rational function

$$\frac{p(x)}{q(x)}, \quad q(x)\neq 0$$

can be expressed as a sum of simpler rational functions, called *partial fractions*. The form of the partial fractions is determined by the factors of the denominator as follows:

1. If $(ax+b)$ is a linear factor of q, include a term

$$\frac{A}{ax+b}.$$

2. If $(ax+b)^N$ with some $N\in\mathbb{N}$ is a repeated linear factor of q, include a term for each power of $(ax+b)$:

$$\frac{A_1}{ax+b}+\frac{A_2}{(ax+b)^2}+\cdots+\frac{A_N}{(ax+b)^N}.$$

3. If ax^2+bx+c is an irreducible quadratic factor of q (quadratics that cannot be factored into real linear factors), include a term

$$\frac{Ax+B}{ax^2+bx+c}.$$

4. If $(ax^2+bx+c)^N$ with some $N\in\mathbb{N}$ is a repeated irreducible quadratic factor of q, where ax^2+bx+c is irreducible over $\mathbb{R}$, include a term for each power:

$$\frac{A_1x+B_1}{ax^2+bx+c}+\frac{A_2x+B_2}{(ax^2+bx+c)^2}+\cdots+\frac{A_Nx+B_N}{(ax^2+bx+c)^N}.$$

The constants appearing in the partial fractions are chosen so that, when the partial fractions are combined, they give the original rational function. They can be determined, for example, by multiplying through by the denominator and then equating coefficients or choosing convenient values of x.

Example 4.25. Find the appropriate partial fraction decomposition of

$$\frac{4x}{x^3 - x^2 - x + 1}.$$

Solution: We first factorise the denominator:

$$x^3 - x^2 - x + 1 = (x-1)^2(x+1).$$

Since the denominator contains the repeated linear factor $(x-1)^2$ and the linear factor $(x+1)$, the partial fraction decomposition has the form

$$\frac{4x}{(x-1)^2(x+1)} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}.$$

Multiplying both sides by $(x-1)^2(x+1)$ gives

$$4x = A(x-1)^2 + B(x+1)(x-1) + C(x+1).$$

Expanding and equating coefficients gives

$$A + B = 0, \quad -2A + C = 4, \quad A - B + C = 0.$$

Solving these equations gives

$$A = -1, \qquad B = 1, \qquad C = 2.$$

Hence,

$$\frac{4x}{x^3 - x^2 - x + 1} = -\frac{1}{x+1} + \frac{1}{x-1} + \frac{2}{(x-1)^2}.$$

□

Example 4.26. Compute $\displaystyle\int \frac{4x}{x^3 - x^2 - x + 1}\mathrm{d}x$.

Solution: From the previous example we have

$$\begin{aligned}\int \frac{4x}{x^3 - x^2 - x + 1}\,\mathrm{d}x &= \int \left[-\frac{1}{x+1} + \frac{1}{x-1} + \frac{2}{(x-1)^2}\right]\mathrm{d}x \\ &= -\ln(|x+1|) + \ln(|x-1|) - \frac{2}{x-1} + C, \quad C \in \mathbb{R},\end{aligned}$$

where we have used that $-1/(x-1)$ is an antiderivative of $1/(x-1)^2$. □

As remarked previously, while the procedure outlined in Result 4.24 only applies when the degree of the numerator is strictly less than the degree of the denominator, it is often possible to first rewrite a rational function as the sum of a polynomial and a rational function for which this condition does hold. For example, we can rewrite the rational function

$$\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1}$$

as

$$\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} = (x+1) + \frac{4x}{x^3 - x^2 - x + 1}.$$

Then, from Example 4.26 we have

$$\begin{aligned}\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1}\,\mathrm{d}x &= \frac{x^2}{2} + x + \int \left[-\frac{1}{x+1} + \frac{1}{x-1} + \frac{2}{(x-1)^2}\right]\mathrm{d}x \\ &= \frac{x^2}{2} + x - \log|x+1| + \log|x-1| - \frac{2}{x-1} + C, \quad C \in \mathbb{R}.\end{aligned}$$

Finally, we remark on an antiderivative that often appears when using the method of partial fractions. From Section 3, we have that

$$\frac{\mathrm{d}}{\mathrm{d}x}\arctan(x) = \frac{1}{1+x^2},$$

which can be used to show that

$$\int \frac{1}{1+x^2}\mathrm{d}x = \arctan(x) + C, \quad C \in \mathbb{R}.$$

Given any $a \in \mathbb{R}$ such that $a \neq 0$, we can then make the substitution $u(x) = x/a$, noting that $u'(x) = 1/a$ so that $\mathrm{d}u = \mathrm{d}x/a$, to obtain

$$\int \frac{1}{a^2 + x^2}\mathrm{d}x = \frac{1}{a}\int \frac{1}{1+u^2}\mathrm{d}u = \frac{1}{a}\arctan(u) + C = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C, \quad C \in \mathbb{R}.$$

Integrals of this form come up very frequently, and therefore this identity can be used to evaluate them.

4.3 Improper integrals

In this final section, we briefly introduce the concept of *improper integrals* in two situations: integrals over unbounded intervals, that is, intervals of the form $(-\infty, a)$ or (a, ∞) that extend indefinitely in at least one direction, and integrals over closed intervals $[a, b]$ where the integrand is undefined or discontinuous at one or more points.

4.3.1 Improper integrals over unbounded intervals

The definite integration we have seen until now is concerned with the signed area under a continuous function defined over bounded intervals. It is possible, however, for the area beneath a function on an unbounded domain to be finite. For example, if $A(t)$ denotes the area beneath the graph of the function $f : [1, t] \to \mathbb{R}$ with $t > 1$ given by $f(x) = 1/x^2$, then

$$A(t) = \int_1^t \frac{1}{x^2}\,\mathrm{d}x = 1 - \frac{1}{t}$$

and we observe that

$$\lim_{t\to\infty} A(t) = \lim_{t\to\infty}\left[1 - \frac{1}{t}\right] = 1.$$

In this way, we interpret the area beneath f on the unbounded interval $[1, \infty)$ as a limit of a family of definite integrals on bounded intervals $[1, t]$. The limit is finite in this particular example because $1/x^2$ decays sufficiently fast as x approaches infinity. This motivates the following definition of improper integrals for functions on semi-infinite domains of the form $[a, \infty)$ and $(-\infty, b]$, as well as on $\mathbb{R}$.

Definition 4.27. The following improper integrals are called *convergent* when the corresponding limits exist in $\mathbb{R}$ (otherwise they are called *divergent*):

- If $f : [a, \infty) \to \mathbb{R}$ is well-defined and continuous on all bounded intervals $[a, t]$, where $t, a \in \mathbb{R}$ with $t > a$, then an improper integral is defined by

  $$\int_a^\infty f(x)\mathrm{d}x = \lim_{t\to\infty}\left(\int_a^t f(x)\mathrm{d}x\right)$$

  whenever the limit exists.

- If $f : (-\infty, b] \to \mathbb{R}$ is well-defined and continuous on all bounded intervals $[t, b]$, where $t, b \in \mathbb{R}$ with $t < b$, then an improper integral is defined by

  $$\int_{-\infty}^b f(x)\mathrm{d}x = \lim_{t\to-\infty}\left(\int_t^b f(x)\mathrm{d}x\right)$$

  whenever the limit exists.

- If $f : \mathbb{R} \to \mathbb{R}$ is continuous and there exists $c \in \mathbb{R}$ such that both improper integrals

  $$\int_{-\infty}^c f(x)\mathrm{d}x \quad \text{and} \quad \int_c^\infty f(x)\mathrm{d}x$$

  are convergent in the above sense, then an improper integral is defined by

  $$\int_{-\infty}^\infty f(x)\mathrm{d}x = \int_{-\infty}^c f(x)\mathrm{d}x + \int_c^\infty f(x)\mathrm{d}x.$$

The following examples show how to calculate these types of improper integral. The main point is to work in two stages, first computing the integral on a bounded domain, and then taking an appropriate limit.

Example 4.28. Consider the following improper integral:

$$\int_1^\infty \frac{1}{x}\,\mathrm{d}x.$$

We calculate

$$\begin{aligned}\int_1^\infty \frac{1}{x}\,\mathrm{d}x &= \lim_{t\to\infty}\left(\int_1^t \frac{1}{x}\,\mathrm{d}x\right)\\ &= \lim_{t\to\infty}\big[\ln(x)\big]_1^t\\ &= \lim_{t\to\infty}\big[\ln(t)-\ln(1)\big]\\ &= \lim_{t\to\infty}\ln(t).\end{aligned}$$

We use the fact that

$$\lim_{t\to\infty}\ln(t)=\infty.$$

Since the limit does not exist as a finite real number, this improper integral is divergent.

Example 4.29. Consider the following improper integral:

$$\int_{-\infty}^\infty e^{-|x|}\,\mathrm{d}x.$$

We first split the integral into two parts, one part on $(-\infty,0]$ and the other part on $[0,\infty)$, and calculate

$$\int_0^\infty e^{-|x|}\,\mathrm{d}x = \lim_{t\to\infty}\left(\int_0^t e^{-x}\,\mathrm{d}x\right) = \lim_{t\to\infty}(-e^{-t}+1),$$

and

$$\int_{-\infty}^0 e^{-|x|}\,\mathrm{d}x = \lim_{t\to-\infty}\left(\int_t^0 e^{x}\,\mathrm{d}x\right) = \lim_{t\to-\infty}(1-e^{t}).$$

We use the fact that

$$\lim_{t\to\infty}e^{-t}=0 \quad\text{and}\quad \lim_{t\to-\infty}e^{t}=0,$$

and therefore

$$\int_0^\infty e^{-|x|}\,\mathrm{d}x = \lim_{t\to\infty}(-e^{-t}+1)=1,$$

and

$$\int_{-\infty}^0 e^{-|x|}\,\mathrm{d}x = \lim_{t\to-\infty}(1-e^{t})=1.$$

We find both parts are convergent, and therefore the original improper integral is also convergent with

$$\int_{-\infty}^\infty e^{-|x|}\,\mathrm{d}x = \int_{-\infty}^0 e^{-|x|}\,\mathrm{d}x + \int_0^\infty e^{-|x|}\,\mathrm{d}x = 1+1=2.$$

4.3.2 Improper integrals over closed intervals

Our consideration of definite integrals has been restricted to functions that are well-defined and continuous on the interval $[a, b]$. However, a function may not be defined or be discontinuous at certain points within $[a, b]$, and an improper integral can instead be defined on subintervals that exclude those singular points. For example, the function $f(x) = 1/\sqrt{x}$ is not defined on the whole of $[0, 1]$, but it is on $[\delta, 1]$ for all $\delta \in (0, 1]$. Indeed, we have

$$\int_\delta^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x = 2 - 2\sqrt{\delta}$$

for all $\delta \in (0, 1]$ so that

$$\lim_{\delta\to 0^+} \int_\delta^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x = \lim_{\delta\to 0^+} \left(2 - 2\sqrt{\delta}\right) = 2.$$

Notice that in this case we have had to consider the one-sided limits we discussed briefly before.

This motivates the following definition of improper integrals for functions that are not defined or discontinuous at certain points on the closed interval $[a, b]$.

Definition 4.30. The following improper integrals are called *convergent* when the corresponding limits exist in $\mathbb{R}$ (otherwise they are called *divergent*):

- If $f : (a, b] \to \mathbb{R}$ is well-defined and continuous on all intervals $[a + \delta, b] \subset (a, b]$, where $a, b, \delta \in \mathbb{R}$ with $b > a$ and $\delta > 0$, then an improper integral is defined by
$$\int_a^b f(x)\mathrm{d}x = \lim_{\delta\to 0^+} \left(\int_{a+\delta}^b f(x)\mathrm{d}x \right)$$
whenever the limit exists.

- If $f : [a, b) \to \mathbb{R}$ is well-defined and continuous on all intervals $[a, b - \delta] \subset [a, b)$, where $a, b, \delta \in \mathbb{R}$ with $b > a$ and $\delta > 0$, then an improper integral is defined by
$$\int_a^b f(x)\mathrm{d}x = \lim_{\delta\to 0^+} \left(\int_a^{b-\delta} f(x)\mathrm{d}x \right)$$
whenever the limit exists.

- If $f : [a, c) \cup (c, b] \to \mathbb{R}$ is well-defined and continuous on all closed subintervals of $[a, c) \cup (c, b]$ for some $c \in (a, b)$, and both improper integrals
$$\int_a^c f(x)\mathrm{d}x \quad \text{and} \quad \int_c^b f(x)\mathrm{d}x$$
are convergent in the sense defined above, then an improper integral is defined by
$$\int_a^b f(x)\mathrm{d}x = \int_a^c f(x)\mathrm{d}x + \int_c^b f(x)\mathrm{d}x.$$

The following examples show how to calculate these types of improper integral.

Example 4.31. Consider the following improper integral:

$$\int_2^5 \frac{1}{\sqrt{x-2}}\,\mathrm{d}x.$$

The integrand is not defined at $x = 2$, so we calculate

$$\begin{aligned}\int_2^5 \frac{1}{\sqrt{x-2}}\,\mathrm{d}x &= \lim_{\delta\to 0^+}\left(\int_{2+\delta}^5 (x-2)^{-1/2}\,\mathrm{d}x\right)\\ &= \lim_{\delta\to 0^+}\left[2(x-2)^{1/2}\right]_{2+\delta}^5\\ &= \lim_{\delta\to 0^+}(2\sqrt{3}-2\sqrt{\delta})\\ &= 2\sqrt{3}.\end{aligned}$$

Example 4.32. Consider the following improper integral:

$$\int_0^1 \frac{1}{x-1}\,\mathrm{d}x.$$

The integrand is not defined at $x = 1$, so we calculate

$$\begin{aligned}\int_0^1 \frac{1}{x-1}\,\mathrm{d}x &= \lim_{\delta\to 0^+}\left(\int_0^{1-\delta} \frac{1}{x-1}\,\mathrm{d}x\right)\\ &= \lim_{\delta\to 0^+}\left[\ln(|x-1|)\right]_0^{1-\delta}\\ &= \lim_{\delta\to 0^+}\ln(\delta).\end{aligned}$$

We use the fact that

$$\lim_{\delta\to 0^+}\ln(\delta) = -\infty.$$

Since the limit does not exist as a finite real number, this improper integral is divergent.

Chapter 5

Extensions and applications

In this final part of the course, we will touch upon some of the many extensions and applications of calculus. One aspect we have referenced many times is the application of calculus to study the real (physical) world, primarily through Newtonian mechanics. We will not discuss this directly here since you will likely see that in another course on differential equations and mechanics. However, many of the extensions and applications outlined here do take their motivation from real-world problems and we will indicate when that is the case.

5.1 L'Hôpital's Rule

The physical interpretation of the derivative we discussed was that of measuring the rate of change in a function given an infinitesimal change in the independent variable. We have also seen that limits describe the behaviour of a function as the independent variable approaches some point or tends to infinity. Suppose we are given two functions, e.g. describing the trajectories of two particles that both tend to infinity as the independent variable tends to infinity: we may then ask, which tends to infinity the "fastest"? In more concrete terms, given $f, g : X \to \mathbb{R}$ with

$$\lim_{x\to\infty} f(x) = \infty, \quad \lim_{x\to\infty} g(x) = \infty$$

what is the correct value of the indeterminate limit

$$\lim_{x\to\infty} \frac{f(x)}{g(x)}$$

of form ∞/∞? Intuitively, we expect this to be zero if g tends to infinity much faster than f, infinity if f tends to infinity much faster than g, or some nonzero number if both functions tend to infinity at roughly the same speed. This principle would hopefully be captured by the derivatives of f and g. Indeed, the following result shows how to rigorously calculate such limits of indeterminant form.

Result 5.1. (L'Hôpital's Rule). Let $(a, b) \subseteq \mathbb{R}$ be an open interval with $a, b \in \mathbb{R}$ and $b > a$ or $a = -\infty$ or $b = +\infty$, and let $x^* \in (a, b)$. Suppose that $f, g : (a, b) \to \mathbb{R}$ are differentiable on $(a, b) \setminus \{x^*\}$ and $g'(x) \neq 0$ on $(a, b) \setminus \{x^*\}$. Moreover, assume that

$$\lim_{x \to x^*} f(x) = \lim_{x \to x^*} g(x) = 0 \tag{5.1}$$

or that

$$\lim_{x \to x^*} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \to x^*} g(x) = \pm\infty. \tag{5.2}$$

(In other words, we have an indeterminate form of type "0/0" or "$\pm\infty/\pm\infty$"). Then

$$\lim_{x \to x^*} \frac{f(x)}{g(x)} = \lim_{x \to x^*} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists, is $-\infty$ or is ∞. Moreover, the same conclusion holds for limits of the form $x \to \pm\infty$, under the analogous hypothesis.

This result immediately allows us to calculate many limits of indeterminant form.

Example 5.2. Find $\lim_{x \to 0} \dfrac{1 - \cos(x)}{x^2}$.

Solution: The functions $f, g : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 1 - \cos(x)$ and $g(x) = x^2$ for all $x \in \mathbb{R}$ are continuous and differentiable on $\mathbb{R}$. Moreover,

$$\lim_{x \to 0} f(x) = \lim_{x \to 0} g(x) = 0$$

and $g'(x) = 2x \neq 0$ whenever $x \neq 0$. Hence, by L'Hôpital's Rule applied to an indeterminant limit of $0/0$ form, we have

$$\lim_{x \to 0} \frac{1 - \cos(x)}{x^2} = \lim_{x \to 0} \frac{(1 - \cos(x))'}{(x^2)'} = \lim_{x \to 0} \frac{\sin(x)}{2x} = \frac{1}{2}$$

where we used the notable limit $\lim_{x \to 0} \sin(x)/x = 1$. □

Example 5.3. Find $\lim_{x \to \infty} \dfrac{e^x}{x^2}$.

Solution: The functions $f, g : \mathbb{R} \to \mathbb{R}$ given by $f(x) = e^x$, $g(x) = x^2$ are continuous and differentiable for all $x \in \mathbb{R}$. Moreover,

$$\lim_{x \to \infty} f(x) = \lim_{x \to \infty} g(x) = \infty$$

and $g'(x) = 2x \neq 0$ whenever $x \neq 0$. Hence, by L'Hôpital's Rule applied to an indeterminant limit of ∞/∞ form, we have

$$\lim_{x\to\infty} \frac{e^x}{x^2} = \lim_{x\to\infty} \frac{e^x}{2x},$$

which is still an indeterminant limit of ∞/∞ form. However, we can again apply L'Hôpital's Rule this time with $f(x) = e^x$ and $g(x) = 2x$ to give

$$\lim_{x\to\infty} \frac{e^x}{x^2} = \lim_{x\to\infty} \frac{e^x}{2x} = \lim_{x\to\infty} \frac{e^x}{2} = \infty.$$

In other words, the function e^x/x^2 tends to infinity as $x \to \infty$. □

The previous example shows that we can continue to apply L'Hôpital's Rule repeatedly, provided at every stage the new quotient still satisfies the relevant hypotheses. Indeed, for any nonzero polynomial $P : \mathbb{R} \to \mathbb{R}$ of degree n we can apply L'Hôpital's Rule n times to show the function $e^x/P(x)$ tends to plus or minus infinity according to the sign of the leading coefficient of P as $x \to \infty$. In this way we understand what it means to rigorously say the exponential function grows much faster than any polynomial.

It is important to remember that L'Hôpital's Rule only works for limits of indeterminate forms $0/0$ or $\pm\infty/\pm\infty$; or other indeterminate forms may sometimes be transformed into one of these forms. For example, consider using the algebra of limits to obtain

$$\lim_{x\to 1} \frac{2x-1}{x} = \frac{\lim_{x\to 1}(2x-1)}{\lim_{x\to 1} x} = \frac{1}{1} = 1,$$

which is not a limit of indeterminate form. However, if we try to apply L'Hôpital's Rule by taking the derivative of the top and bottom, we get a different (incorrect) result:

$$\lim_{x\to 1} \frac{(2x-1)'}{(x)'} = \lim_{x\to 1} \frac{2}{1} = 2.$$

Thus, L'Hôpital's Rule does not apply to determinate forms such as $1/1$; differentiating numerator and denominator in such a case is not generally valid.

Finally, there are certain tricks that enable us to deal with limits of indeterminate form other than $\pm\infty/\pm\infty$ or $0/0$.

Example 5.4. Find $\lim_{x\to\infty} x^{\frac{1}{x}}$.

Solution: This limit is of indeterminate form "∞^0", which does not immediately seem covered by L'Hôpital's Rule. However, we can use the so-called 'exponential trick', by noting that e^x is continuous for all $x \in \mathbb{R}$ so that Result 2.29 on the limits of compositions applies, i.e.

$$\lim_{x\to\infty} x^{\frac{1}{x}} = \lim_{x\to\infty} e^{\frac{\ln(x)}{x}} = \exp\left(\lim_{x\to\infty} \frac{\ln(x)}{x}\right),$$

assuming the exponentiated limit exists. Indeed, the functions $f, g : (0, \infty) \to \mathbb{R}$ given by $f(x) = \ln(x)$, $g(x) = x$ are continuous and differentiable for all $x \in (0, \infty)$. Moreover,

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} g(x) = \infty$$

and $g'(x) = 1 \neq 0$ for all $x \in (0, \infty)$. Hence, by L'Hôpital's Rule applied to an indeterminant limit of ∞/∞ form, we have

$$\lim_{x\to\infty} x^{\frac{1}{x}} = \exp\left(\lim_{x\to\infty} \frac{\ln(x)}{x}\right) = \exp\left(\lim_{x\to\infty} \frac{1/x}{1}\right) = e^0 = 1.$$

□

5.2 Optimisation and curve sketching

Knowing the derivatives of a function $f : X \to \mathbb{R}$ is very useful in applications of calculus to optimisation. For example, if f describes the profit of a company as it varies some input quantity x (e.g. raw material), then it is clearly of interest to find x^* such that $f(x^*)$ is as large as possible. We recall the definition of the maximum and minimum values of a real function $f : X \to \mathbb{R}$.

Definition 5.5. (Global minimum/maximum) Consider a real function $f : X \to \mathbb{R}$ and let $x^* \in X$. Then $f(x^*)$ is the:

- *global maximum* of f on X if $f(x^*) \geq f(x)$ for all $x \in X$;
- *global minimum* of f on X if $f(x^*) \leq f(x)$ for all $x \in X$.

Global maxima/minima are sometimes alternatively called absolute maxima/minima.

The following result gives useful conditions for identifying global maxima and minima of differentiable functions defined on open intervals.

Result 5.6. Let $(a, b) \subseteq \mathbb{R}$ be an open interval with $a, b \in \mathbb{R}$ and $b > a$ or $a = -\infty$ or $b = +\infty$, and let $x^* \in (a, b)$. Suppose that $f : (a, b) \to \mathbb{R}$ is differentiable for all $x \in (a, b)$ and has a global maximum or minimum value at some $x^* \in (a, b)$. Then $f'(x^*) = 0$. Moreover:

- if $f'(x) > 0$ for all $x \in (a, x^*)$ and $f'(x) < 0$ for all $x \in (x^*, b)$ then $f(x^*)$ is the global maximum value of f;

- if $f'(x) < 0$ for all $x \in (a, x^*)$ and $f'(x) > 0$ for all $x \in (x^*, b)$ then $f(x^*)$ is the global minimum value of f.

Example 5.7. The function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ is differentiable for all $x \in \mathbb{R}$ with $f'(0) = 0$. Moreover, $f'(x) = 2x$ is such that $f'(x) = 2x < 0$ for all $x \in (-\infty, 0)$ and $f'(x) = 2x > 0$ for all $x \in (0, \infty)$ so we see that $f(0) = 0$ is the global minimum value of f.

In more general cases, there may be points $x^* \in (a, b)$ where either $f'(x^*) = 0$ or the derivative $f'(x^*)$ does not exist. Points $x^* \in (a, b)$ where either $f'(x^*)$ does not exist or where $f'(x^*) = 0$ are called *critical points* of the function f. A critical point x^* with $f'(x^*) = 0$ is called a *stationary point* of f.

For a continuous function $f : [a, b] \to \mathbb{R}$ that is differentiable on (a, b), any global maximum/minimum is attained either at an endpoint of the closed interval $[a, b]$, or at a point $x^* \in (a, b)$ satisfying $f'(x^*) = 0$ (a stationary point). Thus, the following procedure is used to find global maxima and minima of continuous functions $f : [a, b] \to \mathbb{R}$ that are differentiable on (a, b):

1. Find the values of f at the stationary points of f in (a, b).
2. Find the values of f at the endpoints of the interval, i.e. find $f(a)$ and $f(b)$.
3. The largest of the values obtained in Steps (1) and (2) is the global maximum value, and the smallest is the global minimum value.

Example 5.8. Find the global maximum and global minimum values of the function $f : [-\frac{1}{2}, 4] \to \mathbb{R}$ given by

$$f(x) = x^3 - 3x^2 + 1 \quad \text{for all } x \in \left[-\frac{1}{2}, 4\right].$$

Solution: We use the aforementioned procedure.

1. Since
$$f'(x) = 3x^2 - 6x = 3x(x - 2) \quad \forall\, x \in \left[-\tfrac{1}{2}, 4\right], \tag{5.3}$$
it follows that $f'(x)$ exists for all $x \in \left(-\frac{1}{2}, 4\right)$. The stationary points of f occur when $f'(x) = 0$, that is via (5.3), at $x = 0$ and $x = 2$. Notice that each of these stationary points lies in the interval $(-\frac{1}{2}, 4)$. The values of f at its stationary points are
$$f(0) = 1 \quad \text{and} \quad f(2) = -3. \tag{5.4}$$

2. The values of f at the endpoints of the interval are

$$f(-\tfrac{1}{2}) = \frac{1}{8} \quad \text{and} \quad f(4) = 17. \tag{5.5}$$

3. By comparing these four numbers in (5.4) and (5.5), we see that the global maximum value of f is $f(4) = 17$ and the global minimum value of f is $f(2) = -3$.

□

Example 5.9. Let $f : [-1, 1] \to \mathbb{R}$ be given by $f(x) = x$ for all $x \in [-1, 1]$. Then f has its global minimum value at $x = -1$, and its absolute maximum at $x = 1$, but these values are not found as stationary points of f since $f'(x) = 1$ for all $x \in [-1, 1]$.

The following examples show that one has to be more careful when looking for global maxima and minima of non-differentiable functions or functions defined on open intervals.

Example 5.10. Let $f : [-1, 1] \to \mathbb{R}$ be given by $f(x) = |x|$ for all $x \in [-1, 1]$. Then f has its global minimum value at $x = 0$, but that value cannot be found by solving $f'(x) = 0$ because $f'(0)$ does not exist.

Example 5.11. Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x$ for all $x \in \mathbb{R}$. Then $f'(x) = 1$ for all $x \in \mathbb{R}$, so f has no global maximum or minimum.

Example 5.12. Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^3$ for all $x \in \mathbb{R}$. Then $f'(x) = 3x^2$ for all $x \in \mathbb{R}$, and $f'(0) = 0$, with $x = 0$ the only stationary point of f. However, $f(x) = x^3 > 0$ for all $x > 0$ and $f(x) = x^3 < 0$ for all $x < 0$, so $f(0) = 0$ is neither a global maximum nor a global minimum.

The last example above illustrates that stationary points need not correspond to global maxima or minima. In some cases, they may instead correspond to local maxima or local minima, which describe the behaviour of a function in a neighbourhood of a point.

Definition 5.13. (Local minimum/maximum) Let $f : X \to \mathbb{R}$ be a real function and let $x^* \in X$. We say that $f(x^*)$ is a:

- *local maximum* of f if there exists $\delta > 0$ such that $f(x^*) \geq f(x)$ for all $x \in X$ satisfying $|x - x^*| < \delta$;
- *local minimum* of f if there exists $\delta > 0$ such that $f(x^*) \leq f(x)$ for all $x \in X$ satisfying $|x - x^*| < \delta$.

Thus, a global maximum or minimum compares $f(x^*)$ with the values of f on the whole domain, whereas a local maximum or minimum only compares $f(x^*)$ with values of f sufficiently close to x^*. Consequently, a global maximum is always a local maximum, and a global minimum is always a local minimum, but the converse need not hold. Local maxima and minima of a differentiable function can sometimes be determined by evaluating the second derivative of the function at a stationary point. The following result gives a useful test.

Result 5.14. (Second derivative test) Let $(a, b) \subseteq \mathbb{R}$ be an open interval with $a, b \in \mathbb{R}$ and $b > a$ or $a = -\infty$ or $b = +\infty$, and let $x^* \in (a, b)$. Suppose that $f : (a, b) \to \mathbb{R}$ such that $f''(x^*)$ exists and $f'(x^*) = 0$. Then:

- if $f''(x^*) > 0$, then $f(x^*)$ is a local minimum of f;
- if $f''(x^*) < 0$, then $f(x^*)$ is a local maximum of f.

If $f''(x^*) = 0$, this test is inconclusive: $f(x^*)$ may be a local maximum, a local minimum, or neither.

Example 5.15. Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^2$. Then

$$f'(x) = 2x, \qquad f''(x) = 2.$$

Since $f'(0) = 0$ and $f''(0) = 2 > 0$, the second derivative test shows that $f(0) = 0$ is a local minimum value. In fact, as shown earlier in Example 5.7, it is also the global minimum value of f.

Example 5.16. Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^3 - 3x$ for all $x \in \mathbb{R}$. Then

$$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1),$$

so $x = -1$ and $x = 1$ are the stationary points of f. We also have

$$f''(x) = 6x.$$

Since $f''(-1) = -6 < 0$, $f(-1) = 2$ is a local maximum of f. Since $f''(1) = 6 > 0$, $f(1) = -2$ is a local minimum of f. However, since $f(x) \to \pm\infty$ as $x \to \pm\infty$, f has neither a global maximum nor a global minimum.

Example 5.17. Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^3$ for all $x \in \mathbb{R}$. In Example 5.12 we saw that $x = 0$ was the only stationary point of f. However, since $f''(x) = 6x$ we have $f''(0) = 0$, consistent with the fact that the same argument from Example 5.12 also shows $f(0) = 0$ is not a local minimum/maximum either and the second derivative test is therefore inconclusive in this case.

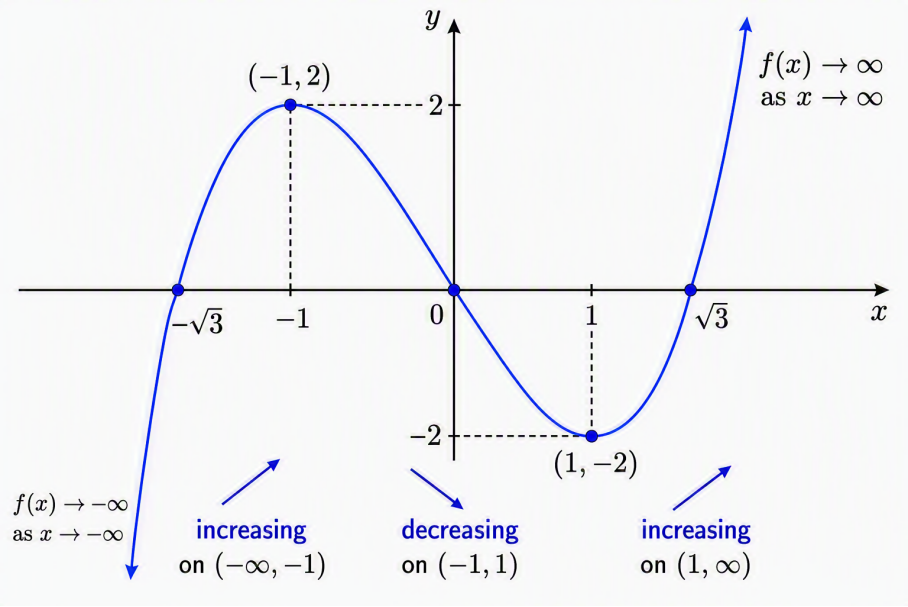


Figure 5.1: Example of curve sketching the function $f(x) = x^3 - 3x$.

Local maxima and minima are important in numerical optimisation because an optimisation algorithm may converge to a local optimum without finding a global optimum. In particular, this motivates the study of classes of functions for which local minima are guaranteed to be global minima. Convex functions have this useful property and therefore play an important role in convex optimisation, which you may encounter later in your degree.

Finally, we provide an example illustrating how the knowledge of derivatives developed above can be combined to sketch the graph of a function.

Example 5.18. We illustrate how the preceding results can be combined to sketch the graph of $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^3 - 3x$. From Example 5.16, the stationary points of f are $x = -1$ and $x = 1$, which correspond to a local maximum at $f(-1) = 2$ and a local minimum at $f(1) = -2$. Moreover, f is increasing on $(-\infty, -1) \cup (1, \infty)$ and decreasing on $(-1, 1)$. We also have

$$f(x) = x(x^2 - 3),$$

so the graph crosses the x-axis at the points $x = -\sqrt{3}$, $x = 0$, and $x = \sqrt{3}$, obtained by solving $f(x) = 0$. Finally, we see that $f(x) \to \pm\infty$ as $x \to \pm\infty$. These observations determine the main features of the graph: it rises to the local maximum at $(-1, 2)$, falls to the local minimum at $(1, -2)$, and then rises again without bound. Together with the three x-intercepts, this information is sufficient to sketch the graph as illustrated in Figure 5.1.

5.3 Multivariable calculus

Up until this section we have only considered calculus involving functions of the form $f : X \to \mathbb{R}$, where the domain $X \subseteq \mathbb{R}$ is some subset of the real numbers, and hence we say f is a function of a single variable $x \in X$. Later in your degree you will likely encounter functions of multiple variables, so-called *multivariable functions* of the form $f : X \to \mathbb{R}$ where the domain $X \subseteq \mathbb{R}^d$ is a subset of d-dimensional real space $\mathbb{R}^d = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$ (d times). By now you should have already seen *vectors* and typically we use vector notation (bold font) to write the output of $f : X \to \mathbb{R}$ at the point $\mathbf{x} \in X$ as $f(\mathbf{x})$. The calculus of functions of a single variable extends to calculus of multivariable functions in many cases, which is called *multivariable calculus*. We will only provide the very basic notions and concepts here, which should serve as a primer for when you encounter multivariable calculus later in your degree.

For a function f of one variable x, the derivative $f'(x)$ expresses the rate of change of $f(x)$ as x varies. In higher dimension, we can ask how the function value $f(\mathbf{x})$ changes as $\mathbf{x} \in \mathbb{R}^d$ varies *along a particular direction*. Let your chosen direction be along the line given by

$$\mathbf{x}(t) = \mathbf{x}_0 + t\mathbf{u}, \quad t \in \mathbb{R}.$$

Recall, this line passes through the point $\mathbf{x}_0 \in \mathbb{R}^d$ and is parallel to the unit vector $\mathbf{u} \in \mathbb{R}^d$. The rate of change of f in the direction $\mathbf{u}$ is then given the *directional derivative* of f in the direction $\mathbf{u}$ at the point $\mathbf{x}_0$:

$$D_{\mathbf{u}} f(\mathbf{x}_0) = \lim_{h \to 0} \frac{f(\mathbf{x}_0 + h\mathbf{u}) - f(\mathbf{x}_0)}{h} = \frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{x}(t)) \bigg|_{t=0}.$$

Example 5.19. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by $f(x, y) = x^2 + y^2$ for $(x, y) \in \mathbb{R}^2$. We want to calculate the directional derivative at the point $\mathbf{x}_0 = (1, 2) \in \mathbb{R}^2$ in the direction of the unit vector $\mathbf{u} = \frac{1}{\sqrt{2}}(1, 1) \in \mathbb{R}^2$. The line through $\mathbf{x}_0$ in the direction $\mathbf{u}$ is

$$\mathbf{x}(t) = (1, 2) + \frac{t}{\sqrt{2}}(1, 1) = \left(1 + \frac{t}{\sqrt{2}}, 2 + \frac{t}{\sqrt{2}}\right)$$

and we therefore have

$$f(\mathbf{x}(t)) = \left(1 + \frac{t}{\sqrt{2}}\right)^2 + \left(2 + \frac{t}{\sqrt{2}}\right)^2 = t^2 + 3\sqrt{2}t + 5.$$

Differentiating with respect to t gives

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{x}(t)) = 2t + 3\sqrt{2}.$$

and after evaluating at $t = 0$ we obtain

$$D_{\mathbf{u}} f(1, 2) = 3\sqrt{2}.$$

Thus, at the point $(1, 2)$, the function f is increasing at a rate of $3\sqrt{2}$ as we move in the direction $\mathbf{u}$.

Example 5.20. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by $f(x, y) = xy$ for $(x, y) \in \mathbb{R}^2$. We want to calculate the directional derivative at the point $\mathbf{x}_0 = (2, 1) \in \mathbb{R}^2$ in the direction of the unit vector $\mathbf{u} = (1, 0) \in \mathbb{R}^2$. The line through $\mathbf{x}_0$ in the direction $\mathbf{u}$ is

$$\mathbf{x}(t) = (2, 1) + t(1, 0) = (2 + t,\ 1)$$

and we therefore have

$$f(\mathbf{x}(t)) = 2 + t.$$

Differentiating with respect to t and evaluating at $t = 0$ gives

$$D_{\mathbf{u}} f(2, 1) = 1.$$

Thus, at the point $(2, 1)$, the function f is increasing at a rate of 1 as we move in the direction $\mathbf{u}$.

Notice what has happened in this second example: moving in the direction $\mathbf{u} = (1, 0) \in \mathbb{R}^2$ means that we change x while keeping y fixed. Similarly, moving in the direction $\mathbf{u} = (0, 1) \in \mathbb{R}^2$ means that we change y while keeping x fixed. These two particular directions are the rectangular cartesian coordinate directions in $\mathbb{R}^2$, associated with unit basis vectors $(1, 0)$ and $(0, 1)$, respectively. This leads us to an especially important special case of the directional derivative, which we state only in three dimensions using the unit basis vectors $\mathbf{i} = (1, 0, 0), \mathbf{j} = (0, 1, 0)$ and $\mathbf{k} = (0, 0, 1)$, but which extends in the same way to any dimension d.

Definition 5.21. Let $f : \mathbb{R}^3 \to \mathbb{R}$, and $\mathbf{i}, \mathbf{j}, \mathbf{k} \in \mathbb{R}^3$ be the unit basis vectors in $\mathbb{R}^3$. Then the directional derivatives $D_{\mathbf{i}} f(\mathbf{x})$, $D_{\mathbf{j}} f(\mathbf{x})$ and $D_{\mathbf{k}} f(\mathbf{x})$, if they exist at $\mathbf{x} \in \mathbb{R}^3$, are called the *partial derivatives* of f with respect to x, y and z, respectively, at $\mathbf{x}$ and denoted

$$\frac{\partial f}{\partial x}(\mathbf{x}) := D_{\mathbf{i}} f(\mathbf{x}), \quad \frac{\partial f}{\partial y}(\mathbf{x}) := D_{\mathbf{j}} f(\mathbf{x}), \quad \frac{\partial f}{\partial z}(\mathbf{x}) := D_{\mathbf{k}} f(\mathbf{x}).$$

Computing partial derivatives is usually straightforward. Consider, for instance, the case associated with unit basis vector $\mathbf{i}$. By definition of the partial derivative, we have

$$\frac{\partial f}{\partial x}(\mathbf{x}) = D_{\mathbf{i}} f(\mathbf{x}) = \lim_{h \to 0} \frac{f(\mathbf{x} + h(1, 0, 0)) - f(\mathbf{x})}{h} = \lim_{h \to 0} \frac{f(x + h, y, z) - f(x, y, z)}{h}.$$

We observe that the limit in this expression is a one-variable limit: in order to compute $\partial f / \partial x$, which we call *the partial derivative of f with respect to x*, it suffices to view f as a function of the single variable x, treating y, z as constants, and to compute the familiar one-variable derivative of the resulting function of x. The same procedure can be followed for computing $\partial f / \partial y$ or $\partial f / \partial z$, by viewing f as a function of y or z alone and treating the other variables as constants. The same principle extends to multivariable functions $f : \mathbb{R}^d \to \mathbb{R}$ defined on any dimension d.

Example 5.22. Let

$$f(x, y) = e^{x^2+y}.$$

To calculate $\partial f/\partial x$, we treat y as a constant. Then

$$\frac{\partial f}{\partial x}(x, y) = e^y \frac{\mathrm{d}}{\mathrm{d}x} e^{x^2} = e^y 2x e^{x^2} = 2x e^{x^2+y}.$$

To calculate $\partial f/\partial y$, we treat x as a constant. Then

$$\frac{\partial f}{\partial y}(x, y) = e^{x^2} \frac{\mathrm{d}}{\mathrm{d}y} e^y = e^{x^2} e^y = e^{x^2+y}.$$

Example 5.23. Let

$$f(x, y, z) = xy^2z^3.$$

Then

$$\frac{\partial f}{\partial x}(x, y, z) = y^2z^3 \frac{\mathrm{d}}{\mathrm{d}x} x = y^2z^3,$$
$$\frac{\partial f}{\partial y}(x, y, z) = xz^3 \frac{\mathrm{d}}{\mathrm{d}y} y^2 = 2xyz^3,$$

and

$$\frac{\partial f}{\partial z}(x, y, z) = xy^2 \frac{\mathrm{d}}{\mathrm{d}z} z^3 = 3xy^2z^2.$$

Finally, we remark there are several notions of integration that are defined for multivariable functions. In multiple dimensions, functions of multiple variables can be integrated over objects (formally, subsets of their domain) on multiple dimensions. *Line integrals* are integrals where the function to be integrated is evaluated along a curve (a one-dimensional object). To make use of integration machinery developed previously, we restrict ourselves to cases where a curve $C \subset \mathbb{R}^2$ in a two-dimensional domain such as $\mathbb{R}^2$ is parameterised by a function $\mathbf{r} : [a, b] \to \mathbb{R}^2$ of the form $\mathbf{r}(t) = (x(t), y(t))$ where $a, b \in \mathbb{R}$ with $b > a$ and $x, y : [a, b] \to \mathbb{R}$ are continuous real functions that are differentiable on (a, b). Curves in higher dimensions can be defined similarly and extended to more general cases (e.g. non-differentiable, non-continuous), but we will not do so here.

To properly define the line integral we introduce notation $\mathbf{r}'(t)$, which here is understood to be the vector of derivatives of x and y evaluated at the point $t \in (a, b)$, i.e.

$$\mathbf{r}'(t) = \left(\frac{\mathrm{d}x}{\mathrm{d}t}(t), \frac{\mathrm{d}y}{\mathrm{d}t}(t) \right),$$

and then using $|\mathbf{r}'(t)|$ to denote the magnitude of the vector $\mathbf{r}'(t)$ we have

$$|\mathbf{r}'(t)| = \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}(t) \right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}(t) \right)^2}.$$

The line integral of a real multivariable function $f : \mathbb{R}^2 \to \mathbb{R}$ over the curve $C \subset \mathbb{R}^2$ parameterised in this way is then defined to be

$$\int_C f \mathrm{d}s := \int_a^b f(\mathbf{r}(t))|\mathbf{r}'(t)|\mathrm{d}t$$

where the expression on the left-hand side can be thought of simply as notation saying "f is integrated over the curve C" and the right-hand side defines concretely what this means mathematically. We can substitute in the expressions of the integrand, which results in a definite integral that we know how to work with.

Example 5.24. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by $f(x, y) = x + y$ for $(x, y) \in \mathbb{R}^2$, and $C \subset \mathbb{R}^2$ be the straight line segment joining $(0, 0)$ to $(1, 2)$ illustrated in Figure 5.2a. We parameterise C by $\mathbf{r} : [0, 1] \to \mathbb{R}^2$ given by $\mathbf{r}(t) = (t, 2t)$ for $0 \leq t \leq 1$. Then

$$\mathbf{r}'(t) = (1, 2) \quad \text{and} \quad |\mathbf{r}'(t)| = \sqrt{1^2 + 2^2} = \sqrt{5}.$$

Also,

$$f(\mathbf{r}(t)) = f(t, 2t) = t + 2t = 3t$$

so that

$$\int_C f \, \mathrm{d}s = \int_0^1 f(\mathbf{r}(t))|\mathbf{r}'(t)| \, \mathrm{d}t = \int_0^1 3t\sqrt{5} \, \mathrm{d}t = \frac{3\sqrt{5}}{2}.$$

Example 5.25. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by $f(x, y) = x$ for $(x, y) \in \mathbb{R}^2$, and $C \subset \mathbb{R}^2$ be the quarter-circle of radius 1 in the first quadrant joining $(1, 0)$ to $(0, 1)$ illustrated in Figure 5.2b. We parameterise C by $\mathbf{r} : [0, \pi/2] \to \mathbb{R}^2$ given by $\mathbf{r}(t) = (\cos(t), \sin(t))$ for $0 \leq t \leq \pi/2$. Then

$$\mathbf{r}'(t) = (-\sin(t), \cos(t)) \quad \text{and} \quad |\mathbf{r}'(t)| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = 1.$$

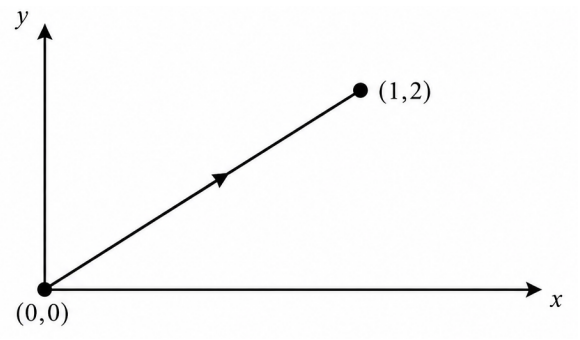
Also,

$$f(\mathbf{r}(t)) = f(\cos(t), \sin(t)) = \cos(t)$$

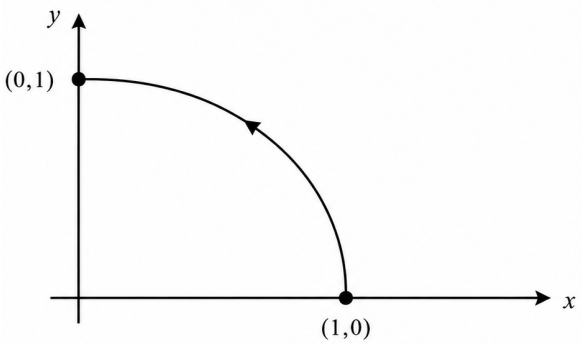
so that

$$\int_C f \, \mathrm{d}s = \int_0^{\pi/2} f(\mathbf{r}(t))|\mathbf{r}'(t)| \, \mathrm{d}t = \int_0^{\pi/2} \cos(t) \, \mathrm{d}t = 1.$$

As remarked previously, line integrals can be defined more generally in any dimension and if you take multivariable calculus you will also see integrals over higher dimensional objects, including surfaces. I hope you look forward to that!



(a) Illustration of curve given in Example 5.24.



(b) Illustration of curve given in Example 5.25.